

Turbulence characterization of downbursts using LES



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ABSTRACT

Loads associated with downbursts represent a significant vulnerability on various structures. Designing the structures to withstand such loads requires the knowledge about the turbulent characteristics of downbursts, which are the focus of the current study. To this effect, large eddy simulations (LES) of downbursts impinging over four different exposures namely open, countryside, suburban and urban, are performed. Ground surface roughness is simulated using fractal surfaces generated by random Fourier modes (RFM) and scaled to match a targeted aerodynamic roughness z_0 . Simulated wind velocities are averaged spatially and temporally to extract the mean and turbulent components. Properties of both the mean and the turbulent components are discussed. Turbulence length scales, which govern the wide band correlations of the turbulence, are determined in the circumferential, the vertical and the longitudinal directions. It is found that the length scales in the circumferential direction are larger than those in the vertical direction by at least an order of magnitude, indicating that downburst turbulence is more correlated in the circumferential direction. This has a particular importance for long horizontal structures such as transmission lines and long span bridges. Narrow band correlations and the turbulent spectra, which have a particular importance for flexible structures, are also discussed. Applicability of using the resulting turbulent characteristics to estimate the peak forces on structures, e.g. transmission lines, is deduced by employing the gust factor approach.

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1. Introduction

Downburst is a strong downdraft that induces an outburst of damaging wind near the ground as defined by Fujita (1985). Hazards associated with downburst winds on different structures are extensively discussed in the literature (Whittingham, 1964; Fujita, 1990; Vicroy, 1992; Holmes, 1999; Li, 2000). Previous field studies such as the Joint Airport Weather Studies (JAWS), the Northern Illinois Meteorological Research on Downbursts (NIM-ROD), and the Federal Aviation Administration Lincoln Laboratory Operational Weather Studies (FLOWS; Fujita, 1985), showed that the maximum downburst wind speeds happen at the 50 m above the ground as indicated by Fujita and Wakimoto (1981), Wilson et al. (1984), and Hjelmfelt (1988). Although field studies can provide the actual velocities, they represent a challenging task due to the unpredictability of the event occurrence in time and in space. That motivated researchers in the past to study downbursts either experimentally (Osegura and Bowles, 1988; Lundgren et al., 1992; Alahyari and Longmire, 1994; Yao and Lundgren, 1996; Wood et al., 2001; Chay and Letchford, 2002) or computationally (Selvam and Holmes, 1992; Hadziabdic, 2005; Chay et al., 2006;

Kim and Hangan, 2007; Sengupta and Sarkar, 2008; Gant, 2009; Mason et al., 2009, 2010a). In terms of the computational studies of downbursts, the following methods are currently used: Impinging Jet (IJ) method proposed by Fujita (1985), Cooling Source (CS) method suggested by Anderson et al. (1992) and the method of simulating the downburst-producing thunderstorm indicated by Orf et al. (2012). Both IJ and CS methods are computationally less costly compared with the simulation of the downburst-producing thunderstorm. The latter requires significant computational resources which makes it unaffordable for the current study.

There are several attempts over the last decades to simulate downburst either using the IJ or the CS methods. For example, Kim and Hangan (2007) used the IJ method to obtain the running mean downburst wind velocities employing an axis-symmetric two-dimensional domain. Sengupta and Sarkar (2008) simulated downbursts using the IJ method employing k-epsilon, k-omega, shear stress transport (SST) and LES turbulence models and compared the resulting profiles with those from an experiment. Their results showed a reasonable agreement between the profiles obtained from the LES and from the experiment. The applicability of using LES to simulate downbursts is also indicated from the results of Hadziabdic (2005), Chay et al. (2006) and Gant (2009). Mason et al. (2009, 2010a) used the CS method to simulate downbursts on a two and three dimensional domains, respectively. Mason et al. (2009, 2010a) used the Shear Adaptive Simulation (SAS) by Menter and Egorov (2005).

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However, SAS model could over-predict the turbulent viscosity of jet-type flows as indicated by Gant (2009). Mason et al. (2009, 2010a) also used the neutral wall function to model the terrain roughness. That was justified because of the small height, Δ_z , of the first grid layer, i.e. 1.0 m according to Teske and Lewellen (1977). Richards and Hoxey (1993), Franke et al. (2004), Fluent Inc. (2005), and Blocken et al. (2007) reported that the physical roughness, k_s or $\sim 30 \cdot z_0$, which can be modeled by a wall function cannot exceed the mid height of the first grid layer, $0.5 \Delta_z$, which leads to the constraint $\Delta_z > 60z_0$. This constraint shades doubts on the results obtained by Mason et al. (2009) for the terrain roughness greater than 0.017 m. Mason et al. (2010b) investigated the effect of the topography on the wind velocities. They estimated the speed-up factors for a downburst and compared them with speed-up factors for synoptic wind. Vermeire et al. (2011a) simulated downbursts over various terrains, with z_0 equals 0.001–0.1 m, using the CS method employing LES to resolve for the turbulence. Similar to Mason et al. (2009), they utilized the neutral wall function with a first grid layer height, Δ_z , of 1.0 m. Later, Vermeire et al. (2011b) used the CS method to study the interaction between multiple downburst events and reported a 55% increase in the velocity magnitude compared to that of a single event. A comparison between the velocity profiles obtained using the IJ and the CS methods is shown in Fig. 1. The profiles obtained by Lin et al. (2007) and Vermeire et al. (2011a) and the instantaneous profile obtained by Mason et al. (2009) using the CS method appear to have maximum velocity close to the ground and quickly drop with height. This could be a result of the different techniques used to enforce the flow in the CS and in the IJ methods. The overall peak profile obtained by Mason et al. (2009b) using the CS method is, however, in a reasonable agreement with those from IJ methods (Vermeire et al., 2011a; Kim and Hangan, 2007). It should be mentioned that in Fig. 1 the velocity profiles generated using the CS method are normalized vertically, assuming the peak velocity happens at a radius equal to $1.2D_{jeq}$, where D_{jeq} is the equivalent diameter for the downdraft formulated by the CS. This allows for a consistent scaling for the data obtained by both the CS and the IJ methods. The choice of $1.2D_{jeq}$ is based on the results by Kim and Hangan (2007).

All of the mentioned above simulations do not discuss the turbulent characteristics (such as turbulence intensities, length scales, spectra and peak factors) of the flow near the ground. These characteristics are essential to quantify the peak loads on different structures and their responses experienced as indicated by Chen and Letchford (2004a, 2004b), Chay and Albermani (2005), Chay et al. (2006), Holmes et al. (2008) and Kwon and Kareem (2009). The current study is an attempt to fill some of these gaps, therefore, focuses on turbulent characteristics of downburst impinging on various exposure conditions. Four exposures namely, open, country side, suburban and urban are considered. Ground roughness corresponding to these exposures is modeled implicitly by using fractal surfaces generated by means of random Fourier modes (RFM) and scaled as necessary represent the targeted aerodynamic roughness of the chosen exposure. Drag forces resulting from the fractal surfaces are then introduced in the flow simulations using the surface gradient drag (SGD) based model originally proposed by Anderson and Meneveau (2010) and latter modified for rougher surfaces by Aboshosha (2014). This model is adopted because (i) it is not bounded by the constraint $\Delta_z > 60 \cdot z_0$ and, therefore allows for modeling rough terrains without losing the accuracy near ground flow simulations where structures engulfed and (ii) it is less computationally demanding compared to explicit roughness element modeling. Simulations are performed in the current study using the IJ method. Although the IJ method does not predict the buoyancy characteristics of the flow as indicated by Vermeire et al. (2011a), it produces an easily scalable wind field as indicated by Shehata et al. (2005) and Kim and Hangan (2007). Generally, the current study is divided into

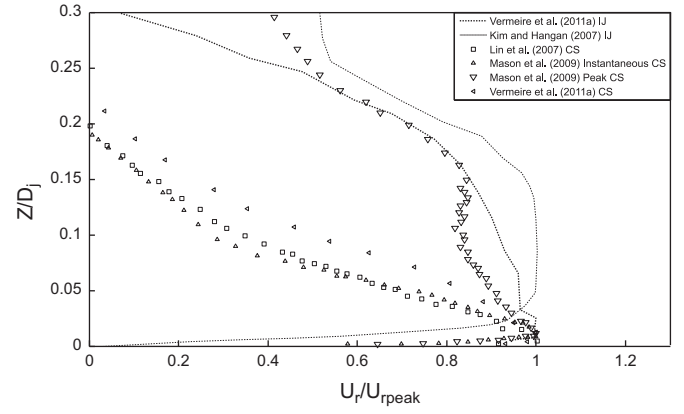


Fig. 1. Comparison between the vertical velocity profile of downbursts using IJ and CS methods.

Table 1

Discretization schemes and solution technique.

Parameter	Type
Time discretization	Second order implicit
Momentum discretization	Bounded central difference
Pressure discretization	Second order
Pressure-velocity coupling	Pressure-implicit with splitting operators (PISO)
Under relaxation factors	0.7 For the momentum and 0.3 for the pressure

four parts: In the first part (Section 2), details of the simulations are provided. In the second part (Section 3), decomposition of the resulting wind field into a mean and a turbulent components is discussed. The third part (Section 4) discusses the simulation results and the findings. The fourth part (Section 5) discusses the application of the downburst characteristics for estimating peak structure responses.

2. LES model setup

The commercial CFD package Fluent (2010) solver is utilized to solve the LES represented by Eq. (1). Dynamic Sub-Grid Scale model by Smagorinsky (1963) and Germano et al. (1991) is used to account for the turbulence. Parameters used to handle flow quantities as well as the solution technique are summarized in Table 1.

$$\begin{aligned}
 \frac{\partial \bar{u}_i}{\partial x_i} &= 0 \\
 \frac{\partial \bar{u}_i}{\partial t} + \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} &= -\frac{1}{\rho} \frac{\partial \bar{P}}{\partial x_i} + \frac{\partial}{\partial x_j} \left(-\tau_{ij} + 2\nu \bar{S}_{ij} \right) + f_i \\
 \tau_{ij} &= \bar{u}_i \bar{u}_j - \bar{u}_i \bar{u}_j \\
 \bar{S}_{ij} &= \frac{1}{2} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \\
 \tau_{ij} - \frac{1}{3} \delta_{ij} \tau_{kk} &= 2\nu_e \bar{S}_{ij} \\
 \nu_e &= (C_s \times \Delta)^2 \left(2\bar{S}_{ij} \bar{S}_{ij} \right)^{1/2}
 \end{aligned} \quad (1)$$

where $i=1, 2, 3$ correspond to the x -, y - and z -directions, respectively. The over bar represents the filtered quantities, u_i , p , t , τ_{ij} and ν represent fluid velocity, pressure, time, the SGS Reynolds stress and molecular viscosity coefficient, respectively. S_{ij} , ν_e , Δ , C_s represent strain rate tensor, eddy viscosity, grid size, Smagorinsky constant which is determined instantaneously based on the dynamic model (Germano et al., 1991), respectively. δ_{ij}

represent Kronecker delta, f_i represent drag force obtained from the modified SGD model (Aboshosha, 2014).

Three dimensional cylindrical domain, illustrated in Fig. 2, is employed to perform the LES. A full cylindrical domain is chosen in the current study compared with only a quadrant domain by Vermeire et al. (2011a). This avoids bounding the flow by the quadrant walls thus allowing evaluation of the turbulent length scales along the circumferential direction. Jet diameter, D_j , is considered equal to 1 km, which represents a typical size of a downburst as indicated by Holmes et al. (2008). The computational domain is chosen to be $8 \cdot D_j \times 4 \cdot D_j$ for the radial and the vertical dimensions, respectively, which is slightly larger than those employed by Vermeire et al. (2011a, b). Two grids, Grid 1 and Grid 2 were used to check the grid independency of the results as shown in Fig. 3 and summarized in Table 2.

In the LES, an instantaneous jet velocity, V_j , of magnitude 40 m/s is used to enforce the flow. This is expected to produce a peak radial velocity in the order of 70 m/s, which is compatible with the maximum velocity recorded during Andrew air-force base downburst event in 1983 (Savory et al., 2001). The results from the

current simulations are applicable for other jet velocities, V_j , following the scaling procedure recommended by Shehata et al. (2005). According to Kim and Hangan (2007), the changes in wind profiles will be negligible at high Reynolds number, Re , adopted in the present study ($Re \sim 10^9$). It should be mentioned that the employed instantaneous jet velocity is expected to have a minor effect on the results. Mason et al. (2009) studied the effect of using a smooth ramp function to enforce the flow considering different smoothing time periods, and their results show a minor effect on the peak velocity and its location for time periods up to 240 s.

In the simulation, a time step of 0.0625 s is chosen to keep Courant Friedrichs–Lewy (CFL) number at the bottom of the computational domain less than one in order to maintain the stability of the solution. Simulations are started from a zero flow condition letting the downburst to develop in the computational domain by the introduced jet. Simulations are performed until the main vortices induced near the inflow by the Helmholtz instability exit the computational domain.

2.1. Simulation of terrain roughness effect

Terrain roughness effect is simulated by using fractal surfaces. Heights of the fractal surfaces, $h(r, \theta)$, are generated according to Eq. (2) using random Fourier modes (RFM).

$$h(r, \theta) = \sum_k S(k) e^{i(kr + \varphi_k)} \quad (2)$$

where $S(k)$ is spectra of the roughness $S(k) = ck^{-(1+\beta/2)}$, k is wave length, c is a constant to control the amplitudes of the fractal surface, β is spectral slope which is taken as equal to -0.5 , φ_k is phase angles $\varphi_k = \tilde{k}_\theta \theta / l_\theta + \varphi_0$; φ_0 represent random phase angles and $\tilde{k}_\theta$ represent Gaussian random numbers with zero mean and 0.5 standard deviation, $1/l_\theta$ represent characteristic wave length in the θ direction which is taken equal to $1/(2\Delta_\theta)$, where Δ_θ is the grid size in the θ direction $= 2\pi/72$.

Heights of the fractal surfaces generated by Eq. (2), need to be scaled so the surface aerodynamic roughness, z_0 , equals to the targeted roughness, z_{0tar} . Scaling is performed using the procedure

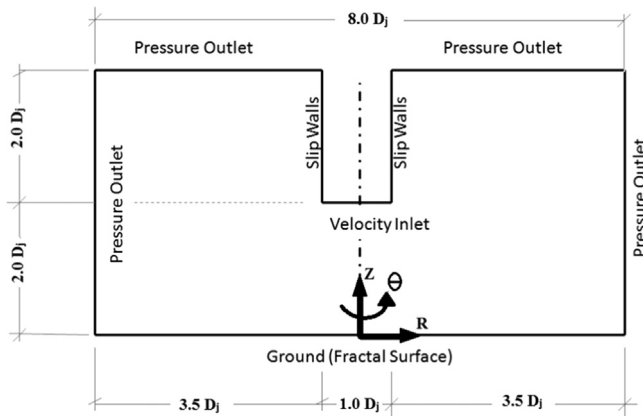


Fig. 2. Computational domain and boundary conditions.

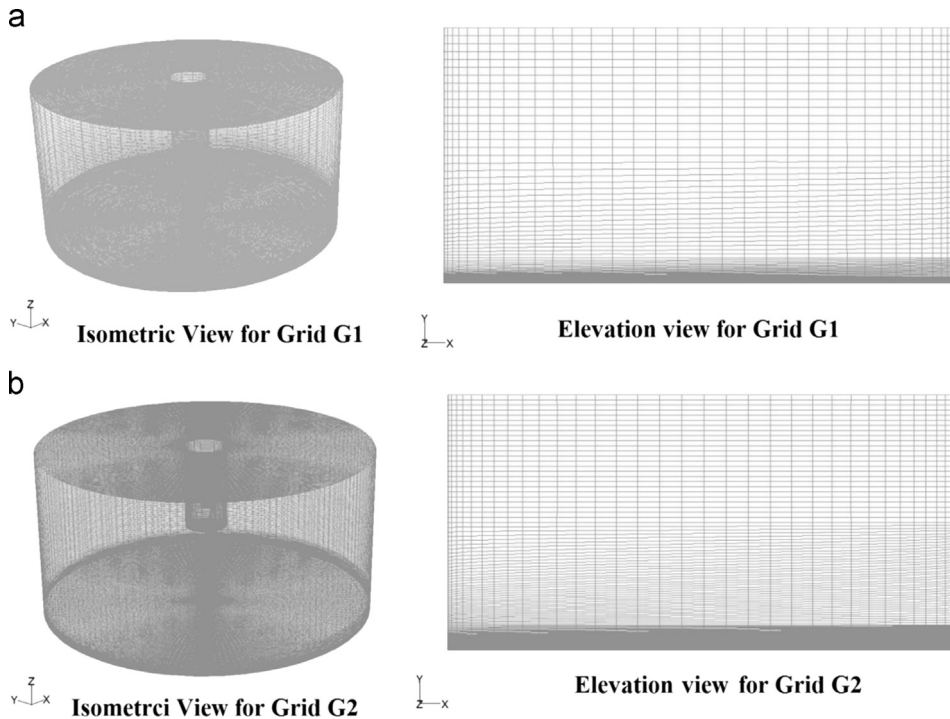


Fig. 3. Employed grids.

Table 2
Properties of the employed grids.

Grid	Grid 1	Grid 2
Radial discretization	400 with $0.01D_j$ each	400 with $0.01D_j$ each
Circumferential discretization	72 with $2\pi/72$ each	72 with $2\pi/72$ each
Vertical discretization	Starts with $0.005D_j$ and increases gradually to $0.10D_j$. Total number of vertical grids is 100	Starts with $0.005D_j$ and increases gradually to $0.07D_j$. Total number of vertical grids is 150
Number of Grids	$2.9E+6$	$4.3E+6$

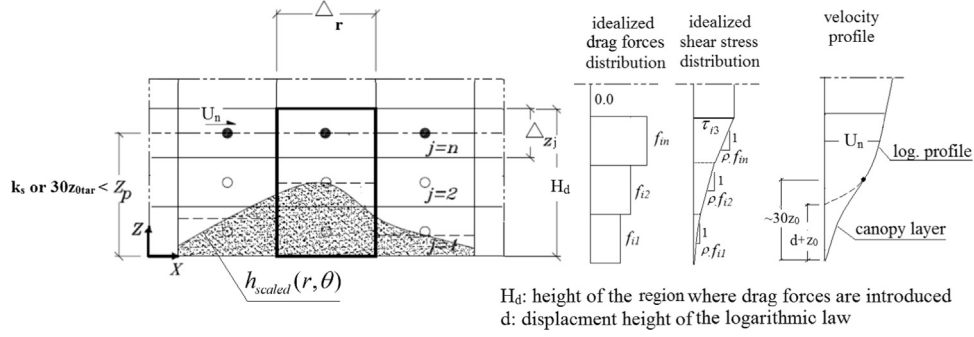


Fig. 4. Illustration of the modified SGD model by Aboshosha (2014).

proposed by Aboshosha (2014) and expressed by Eq. (3), where a is the scaling factor and is expressed by Eq. (4).

$$h_{scaled}(r, \theta) = h_c + ah(r, \theta) \quad (3)$$

where h_c represent a constant height that can be used to set the mean height of the surface to be equal to a specific value, which is chosen in the current study to be half of the physical size of the targeted roughness, $0.5 \cdot k_s - 15 \cdot z_{0tar}$. The constant height does not affect the flow solution, but it affects the overall level of the surface.

$$a = \frac{2}{C_d^* \langle R(\partial h / \partial r) \rangle} \left(\frac{\kappa}{\ln((z_p - d / z_{0tar}))} \right)^2 \quad (4)$$

where C_d^* represent drag coefficient of the roughness elements which relates the drag force to the velocity measured at the reference height z_p , $\langle \dots \rangle$ represent horizontal plane averaging, $R(xx)$ is the ramp function $R(xx) = (xx/2 + |xx|/2)$, κ is von Karman constant and is taken as 0.41, and d represent the displacement height of the logarithmic flow region.

Drag forces resulting from the scaled surface, $h_{scaled}(r, \theta)$, are introduced into the CFD domain using the surface gradient-based drag (SGD) model, proposed originally by Anderson and Meneveau (2010) and modified by Aboshosha (2014). The original model SGD model showed very accurate velocity and Reynolds stress profiles of the flow passing above different surfaces previously examined in the literature (Nakayama and Sakio, 2002; Kanda et al., 2004; Coceal et al., 2007; Xie et al., 2008). The main drawback of the original model is the requirement of placing the physical roughness k_s or $\sim 30 \cdot z_0$ below the centre of the first grid layer, $0.5 \cdot \Delta_z$ (Richards and Hoxey, 1993; Franke et al., 2004; Fluent Inc., 2005; Ansys Ltd., 2005; Blocken et al., 2007). This constraint is the same as that exists in most wall functions and it results from introducing the drag forces in the first grid layer. Aboshosha (2014) modified the SGD model as shown in Fig. 4 by introducing the drag forces into multiple n layers. In the modified model, n can be chosen to place the height z_p , or $n \cdot 0.5 \Delta_z$, in the case of a uniform layer height Δ_z , above the physical size of the roughness elements, k_s or $\sim 30 \cdot z_0$, as illustrated in Fig. 4. This relaxes the constraint on the maximum roughness that can be modeled using a particular grid,

provided that a sufficient number of the layers n is used. In the modified model, shear stress at the top of the layers, τ_{i3} , at the level, H_d , is calculated using Eq. (5), while drag force per unit mass at any layer j , f_{ij} , is expressed by Eq. (6).

$$\tau_{i3} = -\frac{1}{2} \rho C_d^* R \left(\hat{n}_k \frac{\partial h_{scaled}}{\partial x_k} \right) \tilde{u}_{in}^{m\Delta} \quad (5)$$

$$f_{ij} = \frac{\tau_{i3} \tilde{u}_{ij} U_j^{m\Delta}}{\rho \sum_{j=1}^n \tilde{u}_{ij} U_j^{m\Delta} \Delta z_j} \quad (6)$$

where ρ is air density, $\hat{n}_k$ is unit vector of the velocity direction, $\tilde{u}_{in}$ is resolved velocity at the reference height (layer n) in the direction i , $U_n^{m\Delta}$ is magnitude of the velocity at the reference height (layer n) filtered using a filtering width $m\Delta$, where m is calculated according to Aboshosha (2014) as a function of surface heights $h_{scaled}(r, \theta)$, Δz_j is height of layer no. j .

The drag coefficient C_d^* can be evaluated according to Aboshosha (2014) using Eq. (7) as a function of the targeted aerodynamic roughness z_{0tar} .

$$C_d^* = C_d \times \frac{\sum_{j=1}^n \left(\left(I_0(g(z_j)) - I_0(g(z_{01})) / k_0(g(z_{01})) \times k_0(g(z_j)) \right)^2 \times \Delta z_j \right) / \sum_{j=1}^n (\Delta z_j)}{\left(I_0(g(z_p)) - I_0(g(z_{01})) / k_0(g(z_{01})) \times k_0(g(z_p)) \right)^2} \quad (7)$$

$$g(z) = 2 \sqrt{\left(0.768X^2 + 0.9031X \right) \frac{z}{H_d}}$$

$$X = \frac{H_d}{H_d - d} \times \frac{1}{\ln(H_d - d / z_{0tar})} - \left(\frac{H_d - d \ln(H_d - d / z_{0tar})}{H_d - d \ln(H_d / z_{01})} \right)^{1.45} \times \frac{1}{\ln(H_d / z_{01})}$$

where I_0 and k_0 are modified Bessel functions of first and second kinds of zero order, respectively, C_d is the drag coefficient of rectangular-like roughness elements which is equal to 2.0, z_{01} is the aerodynamic roughness due to the surroundings to the roughness elements, which is taken equal to H_d divided by $1e5$ (Wang, 2012).

Expressing the drag forces by Eq. (6) is based on considering an average distribution of the roughness elements in the vertical direction, which is equivalent to the case of roughness elements with the same height but having different shapes in plan (Aboshosha, 2014). In the current study four fractal surfaces are generated using Eqs. (2)–(4) as shown in Fig. 5. The generated surfaces represent open, country side, suburban and urban

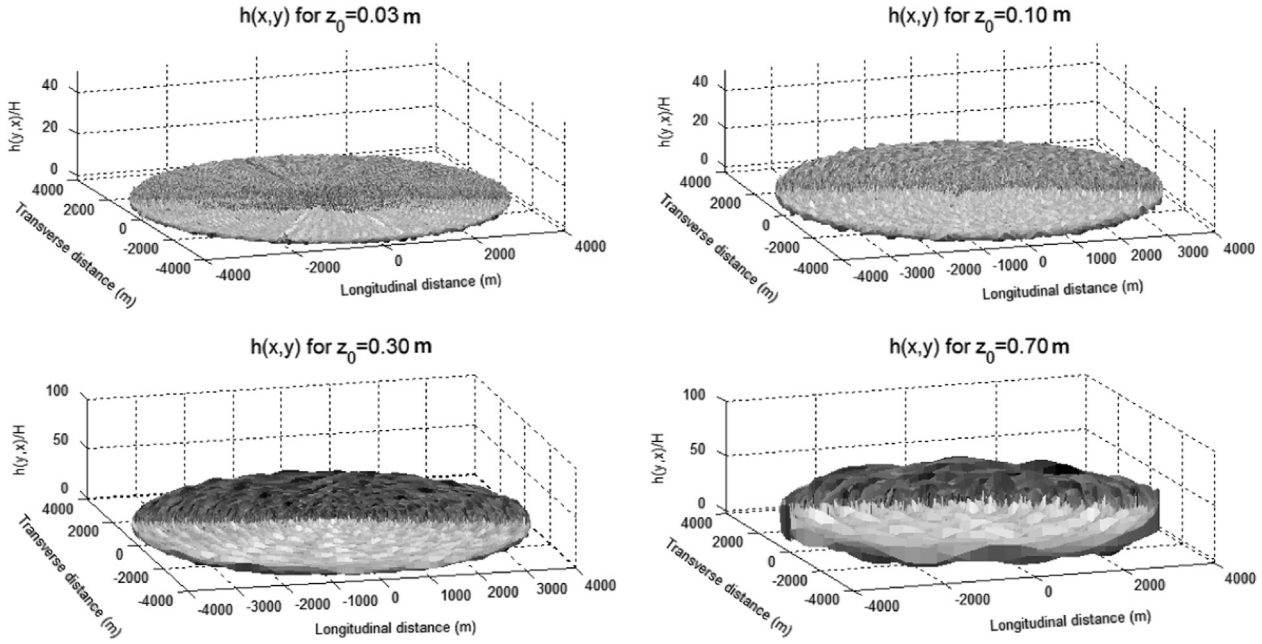


Fig. 5. Roughness produced by using fractal surfaces for four different exposures.

exposures with aerodynamic roughness, z_0 , equal to 0.03, 0.1, 0.3 and 0.7 m, respectively, following the [ESDU \(2001\)](#) definition. Drag forces induced by the fractal surfaces are calculated using the modified SGD model employing Eqs. (5) and (6) and introduced into the governing flow Eq. (1) at a number of grid layers, n , as summarized in [Table 3](#). This number of grid layers is calculated to keep the overall height of the layers where drag forces are introduced, H_d , in the order of $60z_0$ or more.

3. Mean and turbulent wind decomposition

The main difference between decomposing a downburst wind field from synoptic is the time-dependency of its mean component. The wind field is usually decomposed into a running mean and a turbulence component ([Choi and Hidayat, 2002](#); [Holmes et al., 2008](#)). The method adopted for the present study is similar to that used by [Jeong and Hussain \(1995\)](#) to decompose the wind field into a phase average and a random component. A spatial averaging is applied circumferentially at all computational points using a spatial window size having a radial width $dr=0.05D_j$ and vertical height $dz=0.005D_j$, as illustrated in [Fig. 6](#). Resulting velocities from the spatial averaging are also temporally averaged by passing the low frequencies smaller than a cut off frequency f_{cut} . This cut off frequency, given by Eq. (8), is chosen to be twice the shedding frequency, f_{shedd} , of the main vortices. Shedding frequency is taken according to [Kim and Hangan \(2007\)](#) to represent the main vortices happening near the ground at a radius equal to $1.0D_j$.

$$f_{cut} = 2f_{shedd} = 2 \frac{0.3XV_j}{D_j} = 0.6 \frac{V_j}{D_j} \quad (8)$$

where X is the distance from the jet centre to the point of interest which is taken as $2D_j$ to represent the points close to the ground.

Accordingly, a 0.048 Hz cut off frequency, f_{cut} , is used in the current study. This cut off frequency is equivalent to a 69 s averaging period for the real event that happened near Lubbock, Texas, USA in June 2002. This particular event has a jet velocity, V_j , of 29 m/s and a jet diameter, D_j , of 1200 m ([Kim and Hangan, 2007](#)). This is in agreement with a (40–80 s) range recommended

Table 3
Number of grid layers used to introduce the drag forces.

Simulated z_0	0.03 m (Open)	0.1 m (Countryside)	0.3 m (Suburban)	0.7 m (Urban)
H_d (m)	5	10	20	45
n^*	1	2	4	9

H_d : height of the zone where the drag forces are applied. This height is calculated as the round of $60z_0$ to the next grid level (i.e. multiples of 5 m); n is the number of layers where drag forces are applied which is calculated using a fixed layer height that is equal to 5 m (i.e. $0.005D_j$).

for the averaging period by [Holmes et al. \(2008\)](#) and [Darwish et al. \(2010\)](#). [Fig. 7](#) shows the time history of the instantaneous radial velocity U_{r0° located at $R=1.25D_j$, $Z=0.05D_j$ and $\theta=0^\circ$ and the velocity U_{r90° located at $R=1.25 \cdot D_j$, $Z=0.05 \cdot D_j$ and $\theta=90^\circ$. The same figure also shows the resulting time histories after applying the spatial average, U_{rsp} , and after applying both the spatial and the temporal averages, $U_{rsp\&Temp}$. It is clear from the figure that the averaged velocities in the space and time, $U_{rsp\&Temp}$, still contains strong fluctuations similar to those found by [Kim and Hangan \(2007\)](#) for jets with high Reynolds number.

4. LES results and discussions

4.1. Grid independence

Grid independence study is performed for the case of the country side exposure, where z_0 equals to 0.1 m. Maximum averaged velocities in the radial, U_{rmax} , and in the vertical, U_{wmax} , directions are used to check the sensitivity of the results on the employed grids, as illustrated in [Fig. 8](#). Profiles of the radial and vertical velocities obtained from the two grids are in a very good agreement. The maximum difference between the two profiles is found to be 0.83% and 0.98% for the cases of the radial and vertical velocities, respectively. This indicates the independency of the results on the employed grids and therefore, only Grid 1 is used for simulating downbursts on the other exposure conditions. [Fig. 8](#)

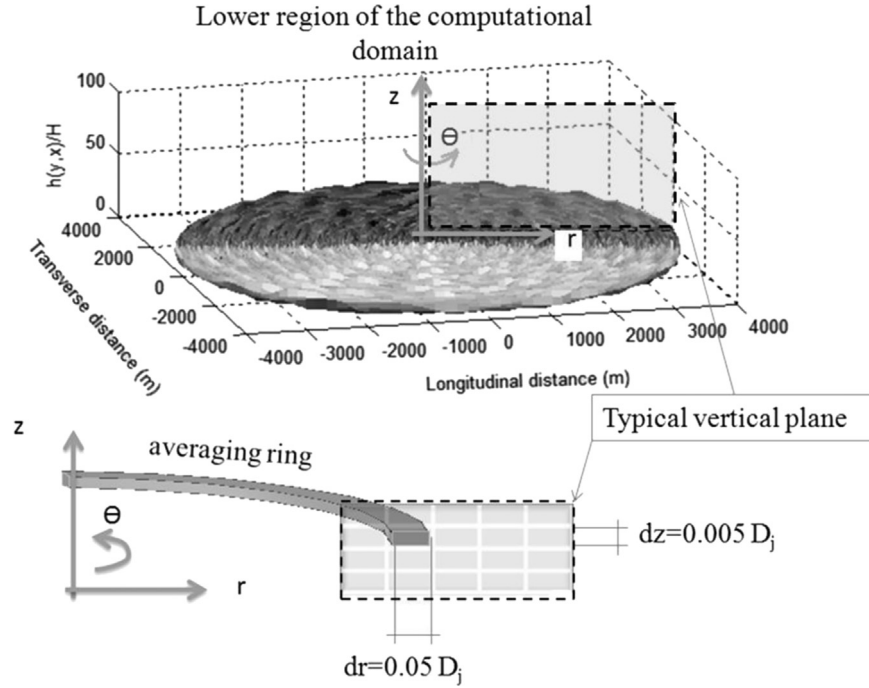


Fig. 6. Spatial averaging of the instantaneous velocities.

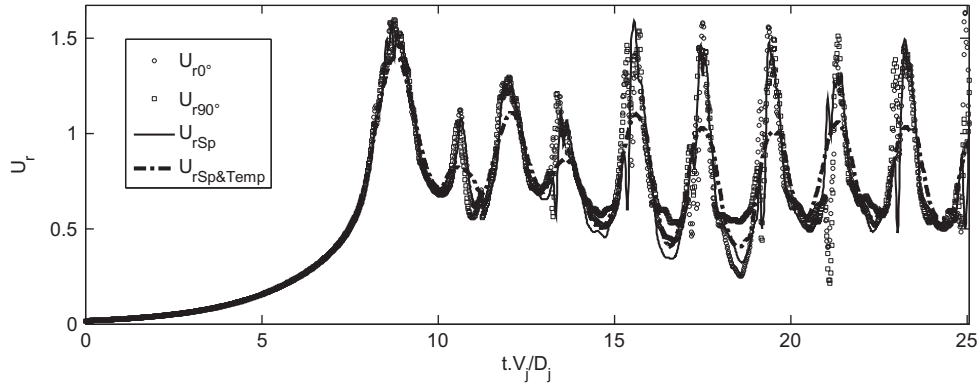
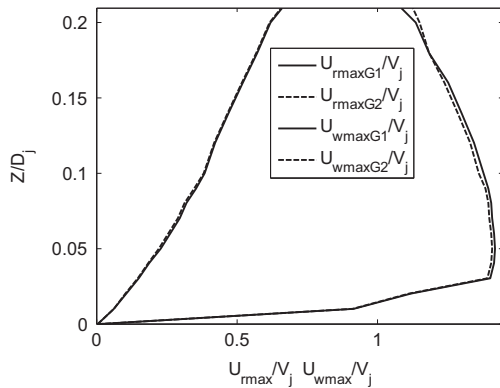
Fig. 7. Spatial and temporal averaging of the instantaneous radial velocity at $R=1.25D_j$ and $Z=0.05D_j$.

Fig. 8. Maximum averaged radial and vertical velocity profiles obtained from Grid 1 (G1) and Grid 2 (G2).

also indicates that the maximum vertical velocities are significantly lower than the maximum radial velocities near the ground, where most structures are located. Therefore, only the radial velocities are discussed in the remaining portion of the paper.

4.1.1. Evolution of the wind field with the time

Downburst is a transient event in which the downdraft impinges towards the ground and convects radially with high velocities. Evolution of the downburst impinging on an open exposure with $z_0=0.03$ m is illustrated in Fig. 9. In this figure, radial velocity and the vorticity contours are plotted at different non dimensional time T_n , where $T_n = \text{Time} \cdot V_j/D_j$. It appears from Fig. 9(a1–a3) that a main vortex is formed right below the velocity inlet boundary due to Helmholtz instability then the vortex travels downward with the jet. After the main vortex hits the ground, as shown in Fig. 9(a4 and a5), it is broken down into multiple smaller vortices that are convected radially. Fig. 9(b1–b5) indicates that high radial velocities are associated with the location of the formed vortices similar to the findings by Kim and Hangan (2007) and Vermeire et al. (2011a).

4.2. Mean wind field

Evolution of the vertical profile of the radial velocity, U_r , for the open terrain condition is illustrated in Fig. 10. Instantaneous vertical profiles are plotted at different radii $R=1.0, 1.5, 2.0$ and $2.5D_j$ from the center. The profiles are plotted at normalized time,



Fig. 9. Evolution of the (a) normalized vorticity and (b) radial velocity for the open terrain condition with the time: (1) $T_n=8$, (2) $T_n=10$, (3) $T_n=12$, (4) $T_n=14$, (5) $T_n=16$; where T_n is the non-dimensional time, which is equal to $Time \cdot D_j / V_j$.

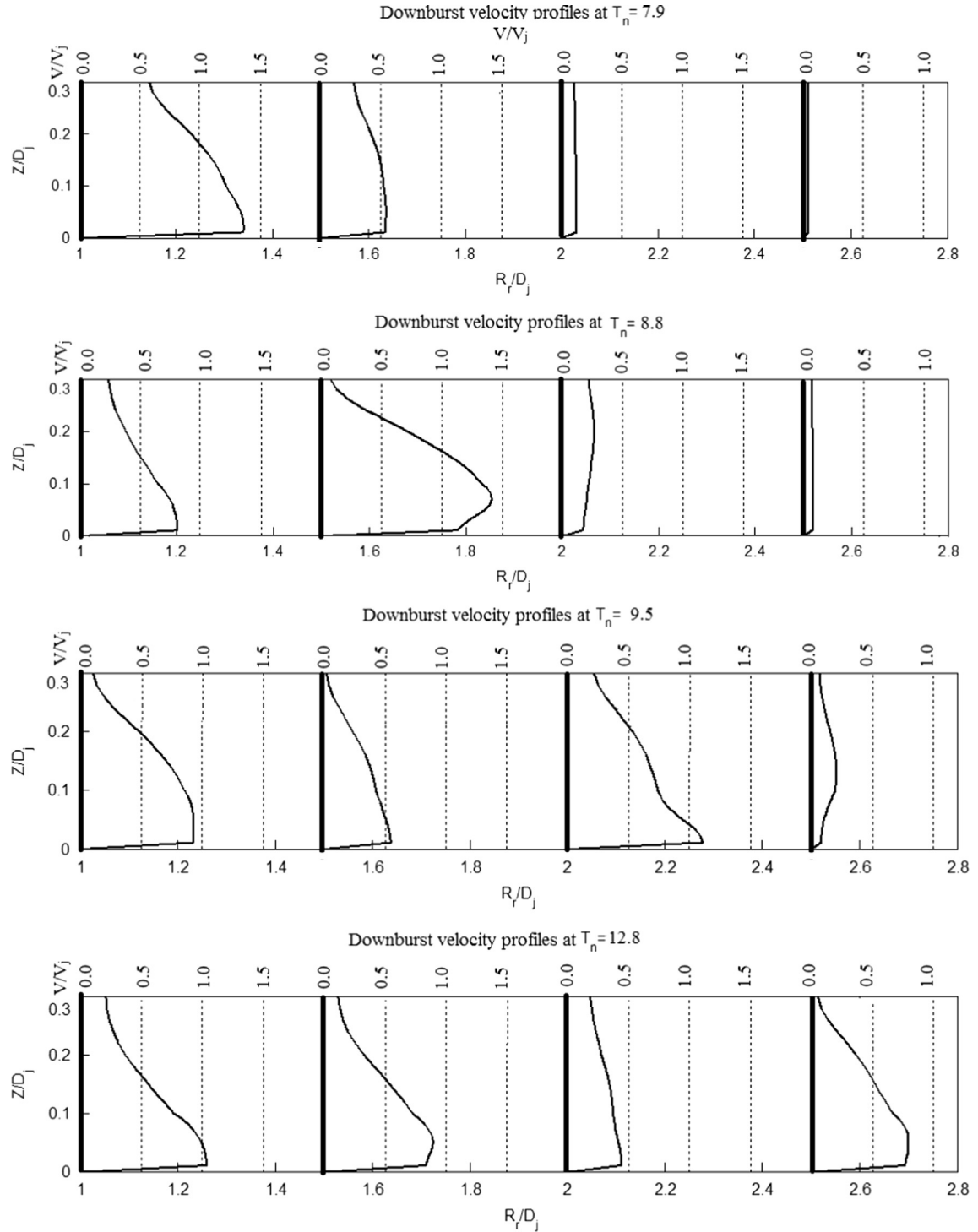


Fig. 10. Radial averaged velocity at $R/D_j = 0.5, 1.0, 1.5$ and 2.0 at $T_n = 7.9, 8.8, 9.5$ and 12.8 .

$T_n = 8.4, 9.2, 11.4$ and 13.0 representing the time instances when the maximum radial velocity occur at those radii ($R = 1.0, 1.5, 2.0$ and $2.5D_j$), respectively.

Profiles of the instantaneous maximum radial velocity and the peak radial velocity (extracted from the entire simulation time) are plotted for the case of the open exposure as shown in Fig. 11. The plotted profiles are normalized by the peak radial velocity, U_{rpeak} , based on the entire computational domain. For the comparison purposes, other profiles obtained from field measurements, experiments and simulations in the literature are also plotted on the same figure.

As shown in Fig. 11, profiles obtained in the current study agree well with the experimental results by Mason and Wood (2004) and Mason et al. (2005), the field measurements by JAWS Data (Hjelmfelt, 1988) and the simulation by Proctor (1988) and Vermeire et al. (2011a). It was observed that both the instantaneous and the peak profiles obtained in the current study to be consistent with the profiles obtained previously. The profile predicted by Kim and Hangan (2007) appears to peak at higher and wider location. This could be attributed to the small scale adopted for their simulations which may have overestimated the thickness of the developed boundary layer (Mason et al., 2009).

It should be mentioned that the present study, due to limitations in the computational resources, adopted a relatively coarse grid far from the ground and fine grids close to the ground as shown in Fig. 3. This has resulted on smoother flow structure at higher altitude for such high Re number flows (i.e. Fig. 9(a3)). To assess the impact of this computational decision, a limited grid independency test was conducted as discussed in Section 4.1 and indicated in Fig. 8. Also, a comparison was made between the profile resulting from the current study and those obtained from experimental and field measurement data as discussed in Section 4.2 and shown in Fig. 11. The obtained profiles show a negligible grid dependency and a good agreement with profiles resulting from other methods. Nevertheless, in the future a high resolution simulation needs to be carried out to assess the effect of this assumption in more detail.

4.2.1. Ground roughness effect

Effect of exposure roughness on the instantaneous maximum radial and the envelope peak (i.e. maximum value of radial velocity at that height at any time) radial velocities are shown in Fig. 12. It appears that both the maximum instantaneous and envelope peak profiles tend to decrease with the increase of the

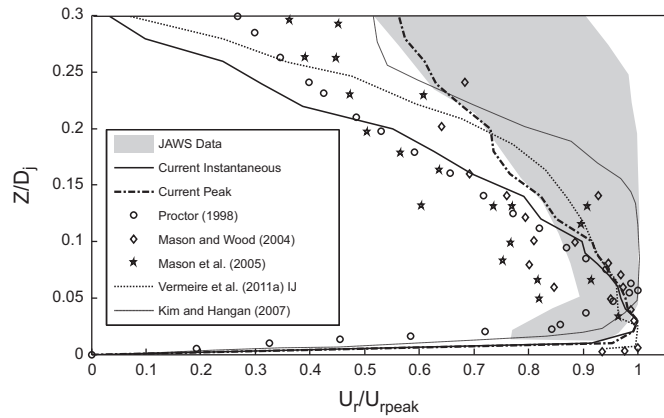


Fig. 11. Radial velocity profile comparisons for the open exposure.

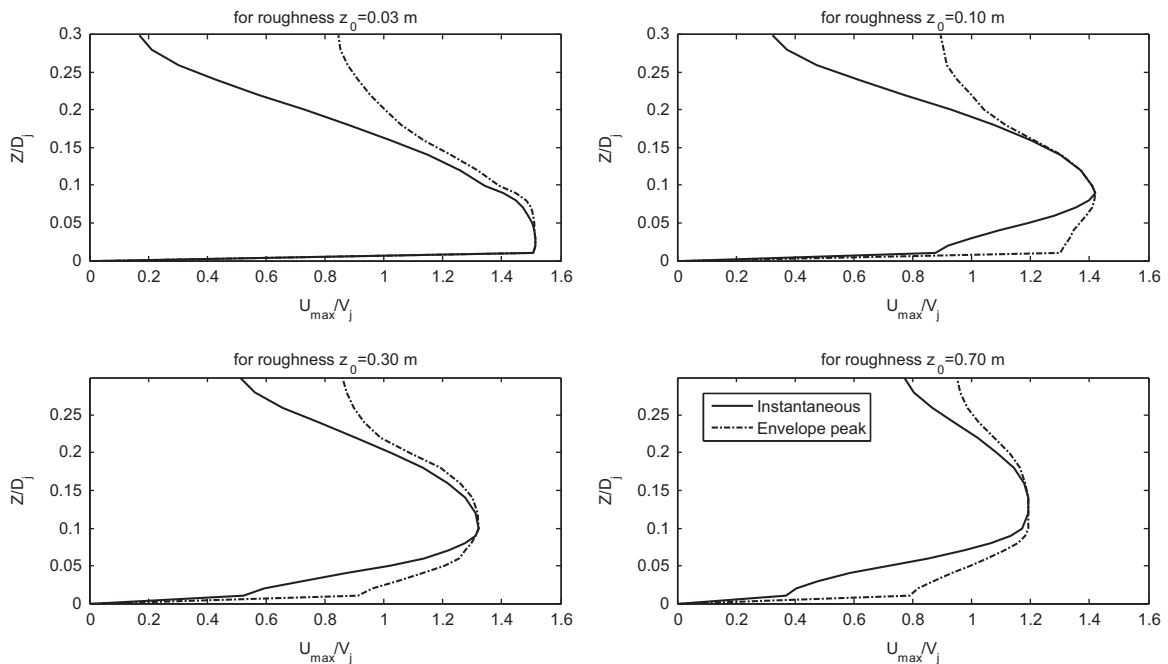


Fig. 12. Instantaneous and envelope peak radial velocity.

exposure roughness. It was also observed that the location of the maximum velocity shifts in the upward direction with the increase of the ground roughness. Fig. 13 shows the contour plots of the peak radial velocities obtained for the four studied exposure conditions. Representative height of the fractal surfaces is marked in the contour plots. Fig. 13 shows that location of the peak radial velocity lays in the range of $R=1.1-1.3D_j$ and tends to increase in z -direction with the increase of the ground roughness as indicated in Fig. 12 as well. The reduction of the peak velocity with increasing the roughness and the location of the peak velocity well agree with the trends reported by Mason et al. (2009) for downbursts enforced by Cooling Sources. By considering a downburst with a diameter of 1000 m (Holmes et al., 2008), the heights of the peak velocity, Z_{max} , are found to be at 30, 90, 100 and 140 m, for $z_0=0.03$, $z_0=0.1$, $z_0=0.3$ and $z_0=0.7$ m, respectively, as shown in Fig. 12. The present values for height of the peak velocity are slightly higher than the values reported by Mason et al. (2009) (i.e. 20 m for $z_0=0.02$ and 50 m for $z_0=0.2$ m). These slight differences can be attributed to the neutral wall function used by Mason et al. (2009) to account for the terrain roughness. As discussed previously, Mason et al. (2009) employed a grid with a first layer height, Δ_z , in the order of 1 m, which can be suitable for open terrain condition (i.e. $z_0=0.02$ m) but may not be suitable for suburban terrain ($z_0=0.2$ m). According to Richards and Hoxey (1993), Franke et al. (2004) and Blocken et al. (2007) the usage of the wall function method is governed by placing the mid height of the first grid layer, $0.5 \Delta_z$, above the physical roughness $k_s \sim 30 z_0$. Based on their recommendations, the maximum aerodynamic roughness z_0 that can be modeled using an order of one meter high first grid is 0.017 m, whereas in the present study the method developed overcomes this limitation for rougher terrains.

4.3. Turbulent wind field

4.3.1. Turbulence intensity

As discussed in Section 3, wind field resulting from the current simulations is decomposed into a mean and a turbulent components. Downbursts are transient events and their mean and turbulent characteristics change with the time. However with

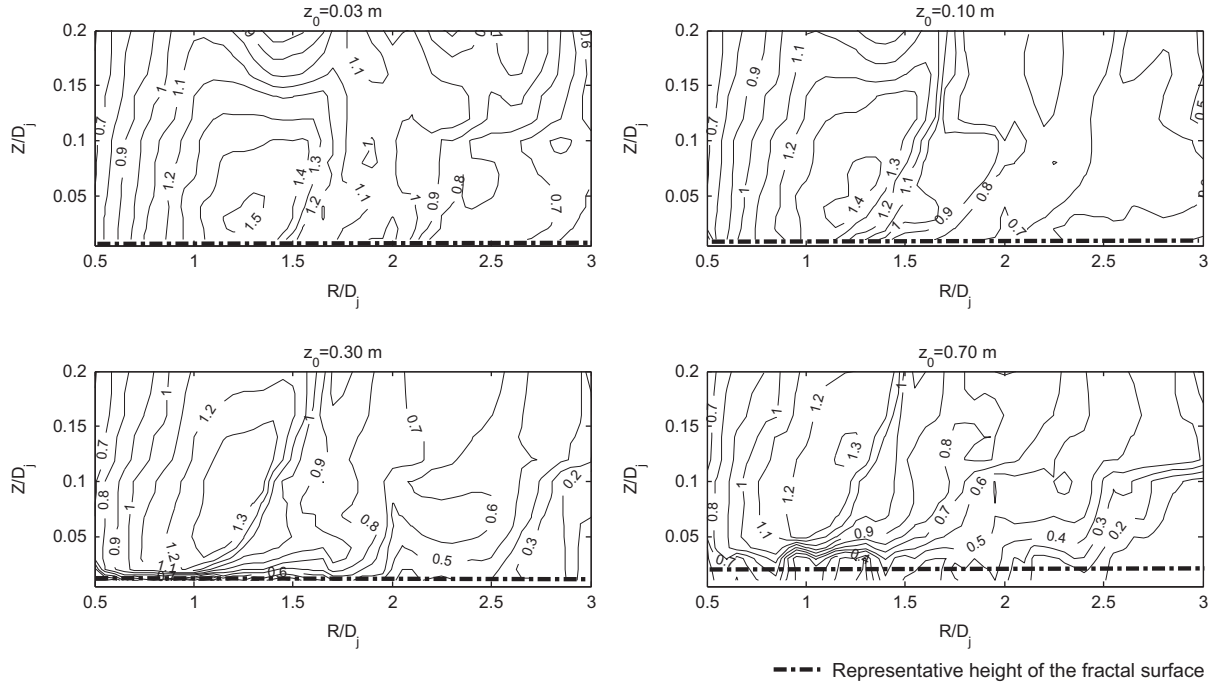


Fig. 13. Maximum mean radial velocity/ V_j .

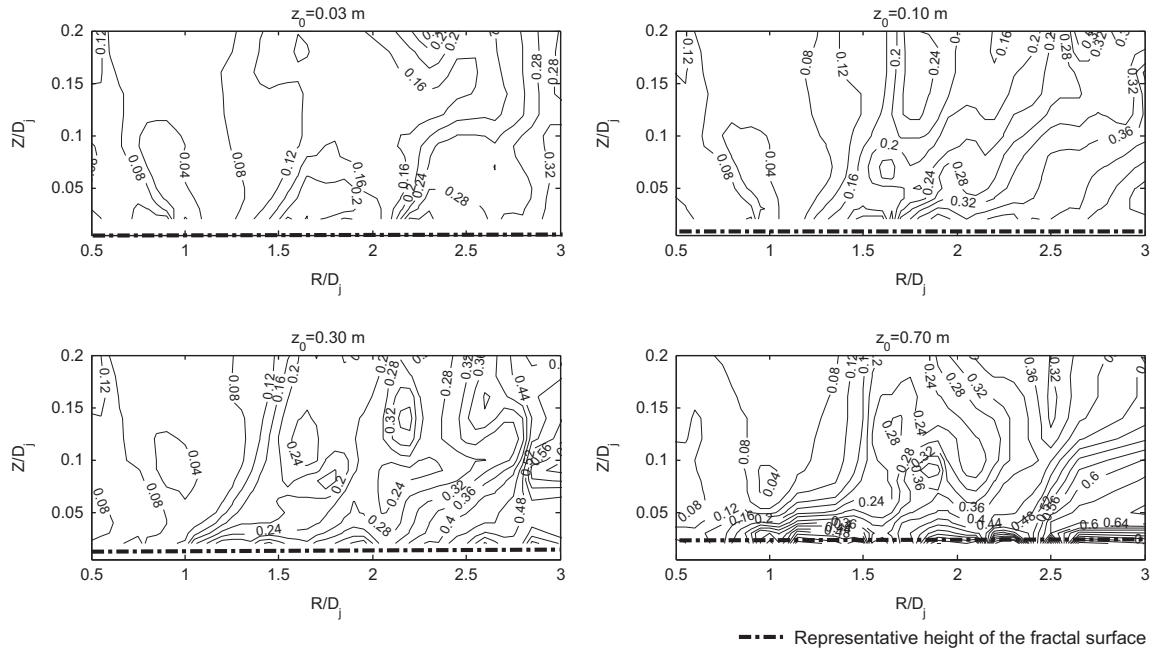


Fig. 14. Turbulent intensity I_{ur} measured at the time instance of the maximum mean radial velocity.

respect to their effects on structures, characteristics near the time instances of the maximum mean velocities are of the main importance as they mostly govern the peak loads associated with the events. Therefore, turbulent characteristics are obtained at those time instances similar to a study reported by Holmes et al. (2008). This is equivalent to treating the downburst turbulence as a piece-wise stationary process and focusing on the time interval close to the time instance of the maximum mean velocities.

Turbulent Intensity, I_{ur} , defined by Eq. (9), is calculated and plotted in Fig. 14 for the four exposure condition considered. The plotted turbulent intensities are based on the resolved fluctuations as the Sub Grid Scale (SGS) contribution to the fluctuations is

found minor (i.e. less than 0.5%). As shown in Fig. 14, turbulent intensity is high near the ground and decreases with the increase in the height. By relating the turbulent intensity obtained from Fig. 14 with the maximum mean velocity obtained from Fig. 13, it is found that the turbulent intensity decreases in the locations where the maximum mean velocity is high. This indicates that the peak velocities are mostly due to the mean component. Turbulent intensity near the ground at the locations of the maximum mean velocities ranges between 0.08 and 0.12, 0.08 and 0.16, 0.08 and 0.24, 0.08 and 0.36, for open, countryside, suburban, and urban exposures, respectively. It is worth noting that the average intensity found in the current study for the open terrain, $I_{ur} \sim 0.10$,

agrees with the findings by Holmes et al. (2008) based on a real downburst event.

$$I_{ur} = \frac{\sigma_{ur \max}}{U_{r \max}} \quad (9)$$

where $U_{r \max}$ is the maximum radial velocity, $\sigma_{ur \max}$ is r.m.s of the fluctuating velocity calculated from the period of $t_{\max} - (1/2f_{cut}) : t_{\max} + (1/2f_{cut})$, where $t_{\max}$ is the time instance corresponding to the maximum mean velocity.

4.3.2. Turbulence correlation in the wide frequency band

Effect of the turbulence on a specific structure is assessed with parameters such as the turbulence intensity, turbulence length scales, and turbulence spectra in addition to the dynamic properties of the structure. In the current study, turbulence length scales in the circumferential, L_θ , the radial, L_r , and the vertical, L_v , directions are evaluated. Circumferential and vertical length scales, L_θ and L_v , are obtained directly, as the fitting parameters, from fitting the spatial correlation functions, $R(d\theta)$ and $R(dz)$, given by Eqs. (10) and (11), respectively.

$$R(d\theta) = \exp\left(-\frac{rd\theta}{L_\theta}\right) \quad (10)$$

$$R(dz) = \exp\left(-\frac{dz}{L_v}\right) \quad (11)$$

where $d\theta$, dz are the magnitude of angular and vertical separations, respectively.

Radial length scales, L_r , can be expressed as a function of the mean velocity and the turbulence time scale, τ_{turb} , as given by Eq. (12). Maximum mean radial velocity, $U_{r \max}$, is used to represent the mean velocity in Eq. (12), as the turbulence is extracted near the time instance of the maximum mean, $t_{\max}$. Turbulent time scale, τ_{turb} , is evaluated by integrating the autocorrelation function, $R(\tau)$, as given by Eq. (13). It should be mentioned that the autocorrelation function may have a negative sign. In such cases, the integration is stopped at the time of the first zero crossing, $T_{0-cross}$, similar to Katul and Parlange (1995).

$$L_r = U_{r \max} \tau_{turb} \quad (12)$$

$$\tau_{turb} = \int_0^{1/f_{cut} \cdot \tau_{0-cross}} R(\tau) d\tau \quad (13)$$

where $\tau_{0-cross}$ is the time corresponding to the first zero crossing of the autocorrelation function $R(\tau)$.

Sample plot of turbulent velocities at a location of $R=1.5D_j$ for the open terrain condition obtained at different angles θ and heights Z are shown in Fig. 15(a), and the correlation functions in the circumferential, the vertical, and the radial directions are plotted in Fig. 15(a–c), respectively. It should be noted that fluctuation with a time scale less than $2\Delta_r/\text{mean}(U_r)$ (i.e. 0.5 s using $\text{mean}(U_r)=40$ m/s) would be filtered out as a result of the grid resolution adopted (i.e. $\Delta_r=0.01D_j$). This grid resolution was selected to make sure at least a 3 s gust that most structures are designed for was captured by the current numerical simulation.

Circumferential correlation function, $R(\theta)$, is calculated using twelve turbulent velocity vectors extracted at every 30° and then fitted with the expression given by Eq. (10), as shown in Fig. 15(b). Vertical correlation function, $R(dz)$, at typical height, Z , is calculated by employing 10 velocity vectors, five on each side of the height extracted at every $0.005D_j$ and then fitted by using Eq. (11), as shown in Fig. 15(c). Autocorrelation function, $R(\tau)$, is calculated by averaging the autocorrelation functions of the velocity vectors taken at every 30° , as shown in Fig. 15(d).

Contours plots of the circumferential length scale, L_θ , are plotted in Fig. 16. It appears from the figure that the circumferential length scale, L_θ , reaches up to 9 times the jet diameter D_j .

This large value indicates that the turbulence associated with downbursts is very well correlated in circumferential direction, which agrees with the findings by Holmes et al. (2008). It is found that large values of, L_θ , cover wider areas in the case of smoother exposures ($z_0=0.03$ and 0.1 m) than the area in the case of rougher exposures ($z_0=0.3$ and 0.7 m). This emphasizes that rougher exposures are able to breakdown the correlated turbulence into a random turbulence. Contour of the vertical length scales, L_v , are plotted in Fig. 17. As shown in the figure, vertical length scales generally ranges between 0.05 and $0.25D_j$, which is relatively smaller compared with the circumferential length scales by at least an order of magnitude. This emphasizes that the turbulence associated with downbursts is less correlated in the vertical direction compared with the circumferential direction, which is favorable in designing tall structures. Turbulent length scale in the vertical direction, L_v , at the location of the maximum mean velocity ($R=1.1-1.3D_j$) is found to be in the order of $0.05 \cdot D_j$. This represents a 50 m length scale for a typical downburst with 1000 m diameter, which is compatible with the length scales defined in the ASCE (2010) for normal wind. Radial length scales, L_r , are plotted in Fig. 18. It appears that a length scale ranges between 0.3 and $0.6D_j$ in the zone where mean velocities are maximum ($R=1.1-1.3D_j$) above the height of the roughness elements. For a typical downburst with 1000 m diameter, this represents a length scale of $300-600$ m which is larger than the longitudinal length scales for synoptic winds $85-150$ m according to the AS/NZS (2011). Larger length scales indicates that the downburst turbulence is better correlated in the wind direction than the turbulence associated with the normal winds. Turbulent length scales, L_θ , L_v and L_r , characterize the correlation in the wide frequency band. This is of a particular importance of quantifying the background forces on structures.

4.3.3. Turbulence spectra

Power spectrum density of the turbulent velocities at four points located at R and Z equal to $(0.02, 1.0) D_j$, $(0.04, 1.0) D_j$, $(0.02, 1.5) D_j$ and $(0.04, 1.5) D_j$ are calculated and plotted in Fig. 19. These points are chosen as they bound the area where the peak velocities are expected. It is worth mentioning that frequencies smaller than f_{cut} are not shown in the figure as they correspond to the mean component. For comparison purposes with the normal wind, Von Karman spectra, represented by Eq. (14) AS/NZS (2011), is also plotted in Fig. 19.

$$S_{ur} = \frac{4\sigma_u^2 (L_r/U_m)}{(1 + 70.8((fL_r/U_m))^2)^{5/6}} \quad (14)$$

where σ_u is the r.m.s fluctuation, U_m is mean velocity, which is taken equal to the maximum mean velocity $U_{r \max}$, f represent frequency, L_r represent turbulent length scale in the radial direction which is taken from Fig. 18.

Fig. 19 shows that the spectra obtained at the radius R equals to $1.0 \cdot D_j$ agree reasonably with von Karman's especially for the rougher exposures. With the increase of the radius, a steeper slope than the $-2/3$ of von Karman is found. This agrees with the finding by Holmes et al. (2008) for a real downburst event although they reported a less steep slope. Generally, the steeper slope indicates that flexible structures with natural frequencies, $0.1-1$ Hz, are less susceptible to the dynamic excitation by downburst turbulence than by the normal wind turbulence. Fig. 19 also shows the roughness effect on the spectra. It is found at the radius R equals to $1.0 \cdot D_j$, that the turbulent fluctuation associated with small eddies is higher for the rough terrains, $z_0=0.3$ and 0.7 m, than the fluctuations for smooth exposures, $z_0=0.03$ and 0.1 m. However, with increasing the distance from the downburst jet, $R=1.5D_j$, energy associated with the smaller eddies becomes

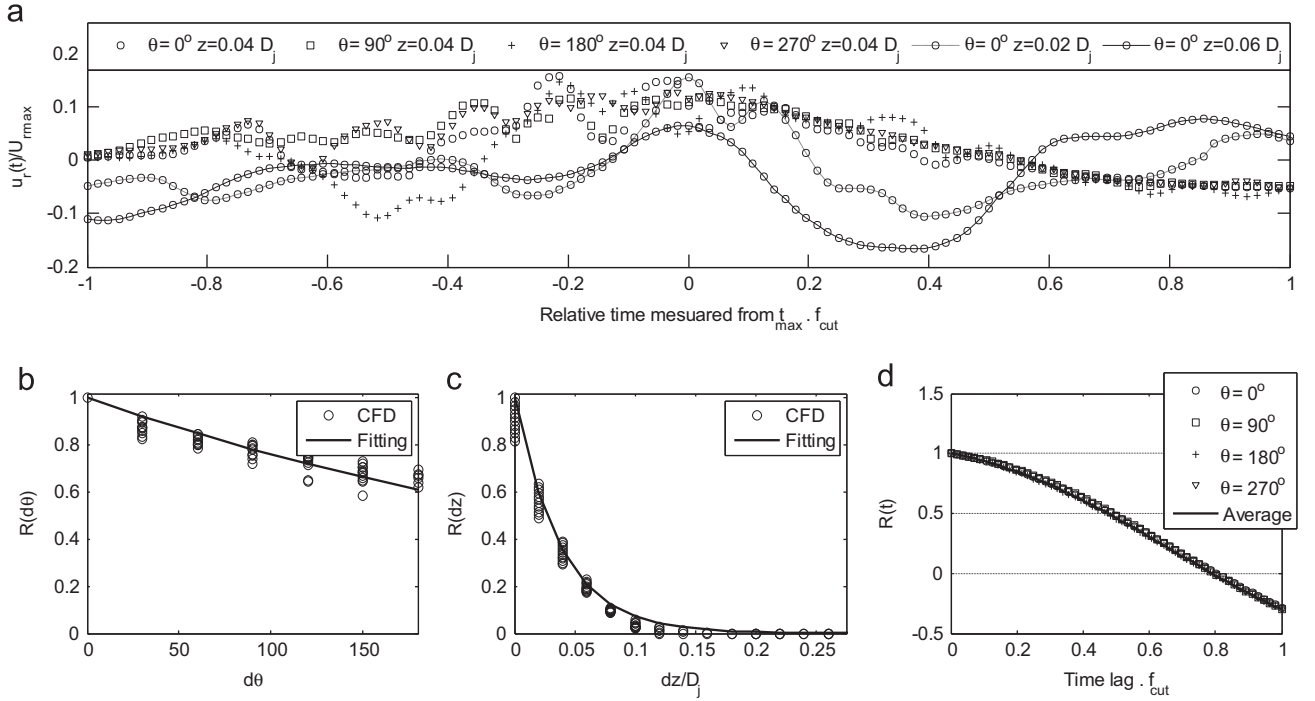


Fig. 15. Procedure of obtaining the length scales from the fluctuating velocities: (a) fluctuating radial velocities, (b) fitting the circumferential correlation function $R(d\theta)$, (c) fitting the vertical correlation function $R(dz)$ and (d) averaging the autocorrelation function.

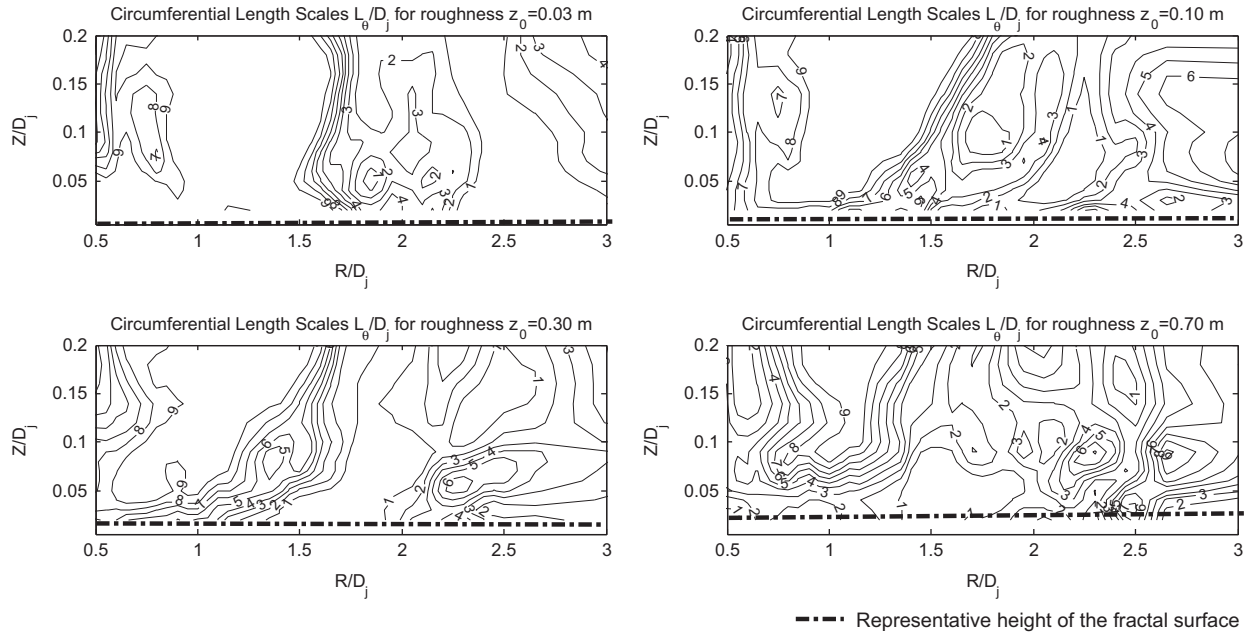


Fig. 16. Circumferential length scale of turbulence L_θ .

higher in the smooth exposures $z_0 = 0.03$ – 0.1 m. This indicates that rough exposures are able to breakdown the turbulence into smaller eddies in shorter distances than smooth exposures.

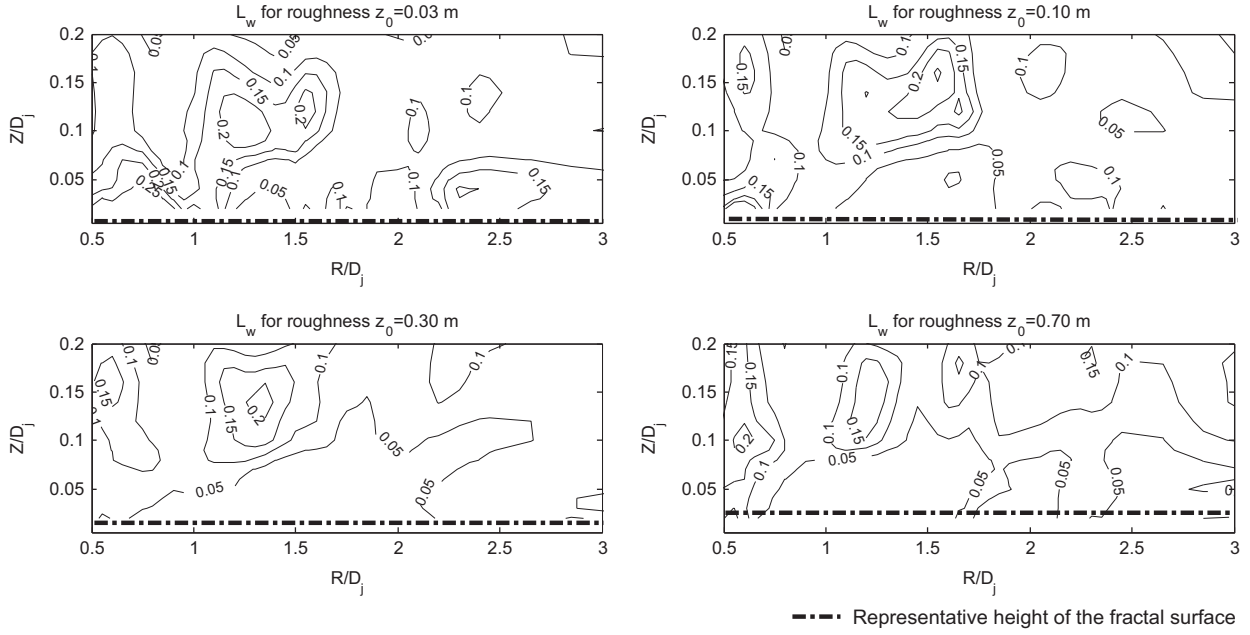
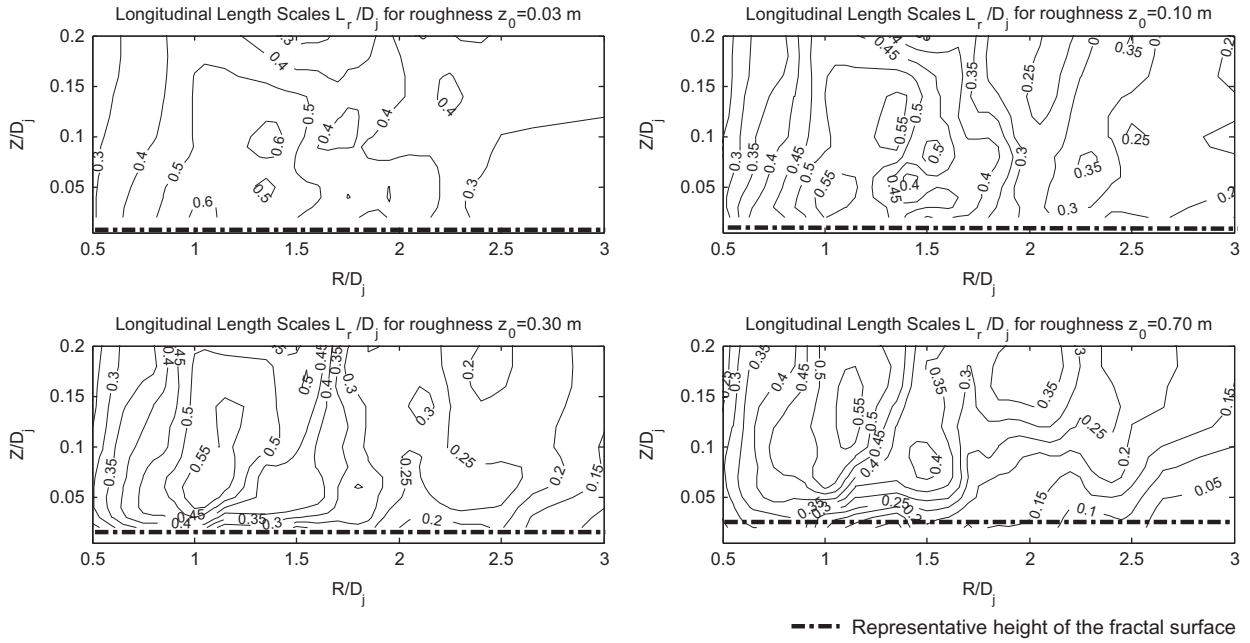
4.3.4. Peak radial wind velocity

Downburst turbulent component is non-stationary meaning that its characteristics change with the time. In the current study, as discussed earlier, turbulent parameters are evaluated at the time instance of the maximum mean velocities. This is equivalent to treating the downburst as a piece-wise stationary process with Gaussian assumption and focusing on the time interval close to the

time instance of the maximum mean velocity. This would help to calculate the peak radial velocity, $\hat{U}_r$, by using Eq. (15). The peak factor, g_v , represents the ratio between the peak fluctuations to the r.m.s fluctuations and can be calculated from Eq. (16). Estimating the peak factor statistically is more stable than the estimation using the absolute peak velocities.

$$\hat{U}_r = U_{r \max} (1 + g_v I_{ur}) \quad (15)$$

$$g_v = \sqrt{2 \ln(2\nu T)} + \frac{0.5772}{\sqrt{2 \ln(2\nu T)}} \quad (16)$$

Fig. 17. Vertical length scales of turbulence L_w .Fig. 18. Radial turbulence length scale L_r .

where ν is the rate of the zero crossing of the fluctuating velocity, T is averaging time which is equal to $1/f_{cut}$.

Average rate of zero crossing ν , which represents the average frequency of the turbulence, is required in order to obtain the peak factor, g_w , using Eq. (16). Expression given by Eq. (17), is used to obtain such an average rate, ν . The resulting peak factor, g_w , is plotted in Fig. 20. As shown in Fig. 20, peak factor, g_w , is less sensitive to the location in the domain and ranges between 2.00 and 2.20.

$$\nu = \sqrt{\frac{\int_{f_{cut}}^{f_s/2} (f - f_{cut})^2 S_u(f) df}{\int_{f_{cut}}^{f_s/2} S_u(f) df}} + f_{cut} \quad (17)$$

where $S_u(f)$: spectra of the radial velocity, f_s : sampling frequency.

4.3.5. Turbulent correlations in the narrow frequency band

Turbulent correlations in the narrow frequency band are required for the cases where the structures are susceptible to the dynamic excitation by the wind turbulence. Turbulent correlations in the narrow frequency band can be represented by the root coherence, $\rho(f)$, given by Eq. (18).

$$\rho(f) = \exp\left(\frac{-Cfd}{U_{max}}\right) \quad (18)$$

where f is frequency, U_{max} is maximum mean velocity, C is coherency decay constant, and d is distance between the velocity pairs, which is taken equal to dz and $Rd\theta$ for the vertical and the circumferential directions, respectively.

The constant C in Eq. (18) is called the coherency decay constant. It characterizes the correlations in the narrow frequency band. High

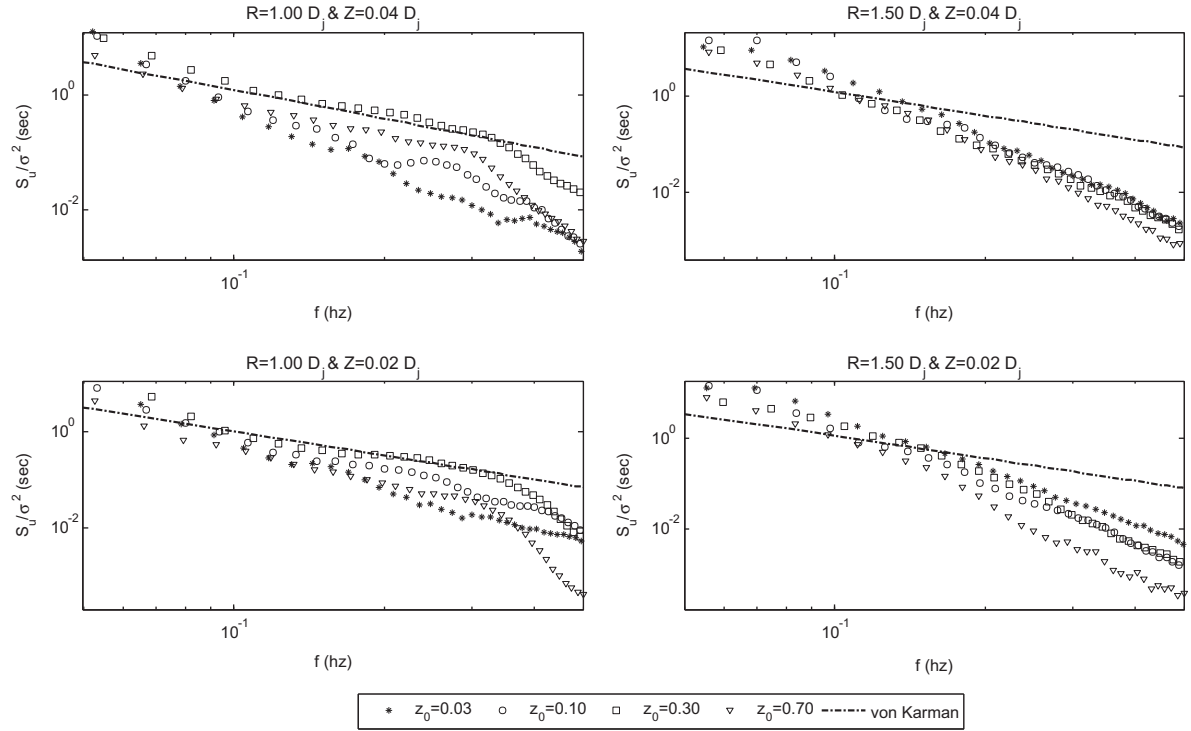
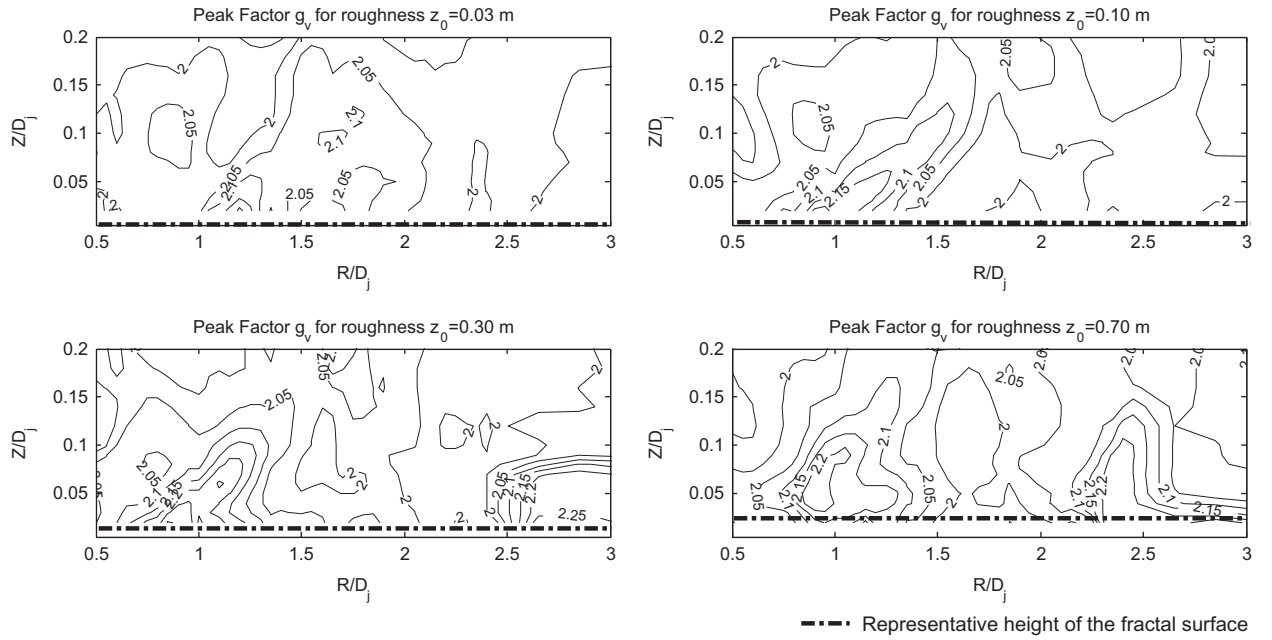


Fig. 19. Turbulent spectra.

Fig. 20. Peak factor g_v .

values of C represent low correlations, while low values indicate high correlations. Variation of the narrow band correlations characterized by the coherency decay constant, C , is studied. Root coherence, $\rho(f)$, is plotted for the turbulent velocity vectors obtained at different vertical and circumferential locations. Sample root coherences for velocity vectors that vary in the vertical and the circumferential directions at coordinates R, Z equal to $(2.0D_j, 0.004D_j)$ are plotted in Fig. 21(a and b), respectively. By fitting the root coherence with the expression in Eq. (18), coherency decay constants in the vertical, C_w , and in the circumferential, C_θ , directions are obtained as the fitting parameter, as shown in Fig. 21(a and b), respectively. Variation of the coherency decay constants, C_w , and, C_θ , is shown by the contour plots

in Figs. 22 and 21, respectively. As shown in these figures, the constant, C_w , generally decreases with the increase of the height, which agrees with the findings by Chen and Letchford (2005). It is found that the constant C_w is of the order of 10 at the location of the maximum mean velocities, i.e. $R=1.1-1.3D_j$. This is consistent with the range of values used for normal winds, 5–15 (Holmes et al., 2008). The decay constant in the circumferential direction, C_θ , has relatively smaller values, which are expected because of the well correlation of the turbulent in the circumferential direction. This means that downbursts can better excite flexible long horizontal structures by a well correlated turbulence with frequencies close to the structures' frequencies than normal winds. Fig. 23

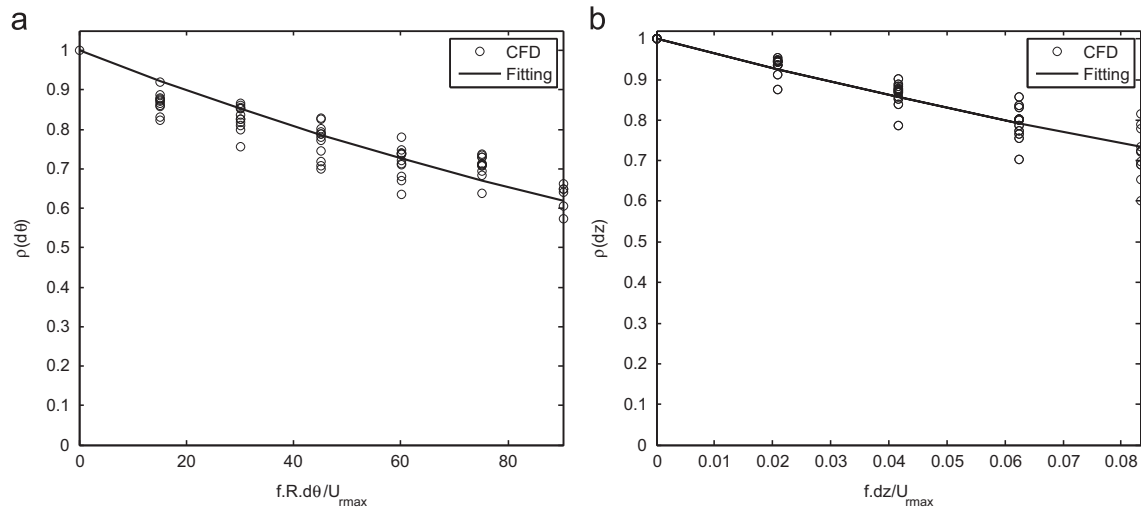
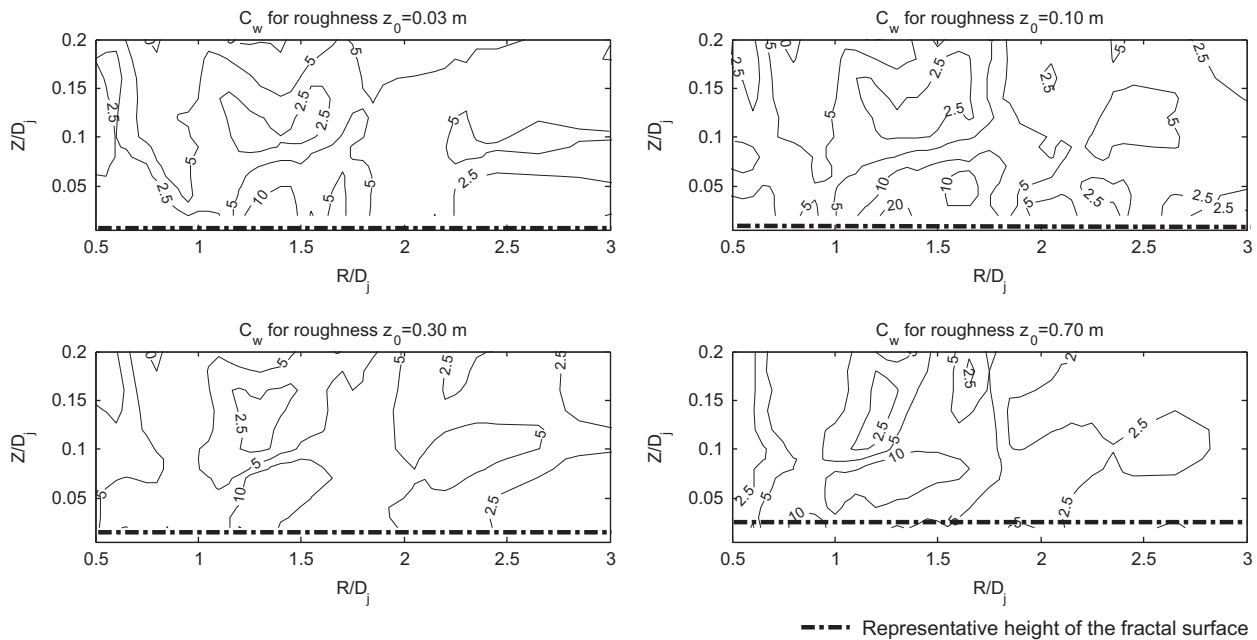


Fig. 21. Fitting the root coherence function.

Fig. 22. Coherency decay constant in the vertical direction C_w .

5. Application for estimating the peak responses of structures

Effective wind load by the gust wind is commonly related to the mean load using the gust factor (GF) approach. Gust factor for a structure, originally proposed by Davenport (1967) represents the ratio between the peak displacement to the mean displacement. Such a GF is extensively examined by Solari (1993a, 1993b) and Simiu and Scanlan (1996). Later Zhou and Kareem (2001) suggested a new definition for the gust factor as the ratio between the peak base moment and the mean base moment. The new definition is believed to allow for a more accurate estimation of the effective forces felt by the structures. Gust factor approach widely used in the design codes, such as the ASCE (2010) and the AS/NZS (2011), is valid for synoptic winds. Kwon and Kareem (2009) suggested a new frame work called the gust front factor (GFF) which is valid for both synoptic and non-synoptic winds including downbursts. The new GFF includes some factors to account for the non-stationarity associated with the non-synoptic winds. Those factors converge to unity for the case of stationary wind letting the new GFF converging to the original GF.

The new GFF involves intensive calculations and, therefore, Kwon and Kareem (2009) proposed a web-based approach to perform those calculations. Although, the web-based approach includes many assumptions on the characteristics of downbursts, it accepts user-defined downburst characteristics and, therefore, allows for evaluating the GFF using more accurate characteristics. It should be mentioned that downburst characteristics obtained from the current study can be implemented through Kwon and Kareem's user-defined option to evaluate the GFF.

It should be mentioned that the new GFF proposed by Kwon and Kareem (2009) is developed to cover dynamically sensitive and insensitive structures. By considering dynamically insensitive structures such as typical transmission lines (Holmes, 2008), calculation of the GFF can be greatly simplified by focusing on the mean and the background components. Considering only the mean and the background components is a common practice in the design codes for typical TL structures under synoptic winds such as the ASCE (2010); IEC (2003) and the AS/NZ (2010). By considering only the mean and the background components, GFF for a response R can be expressed by Eq. (19) using the statistical

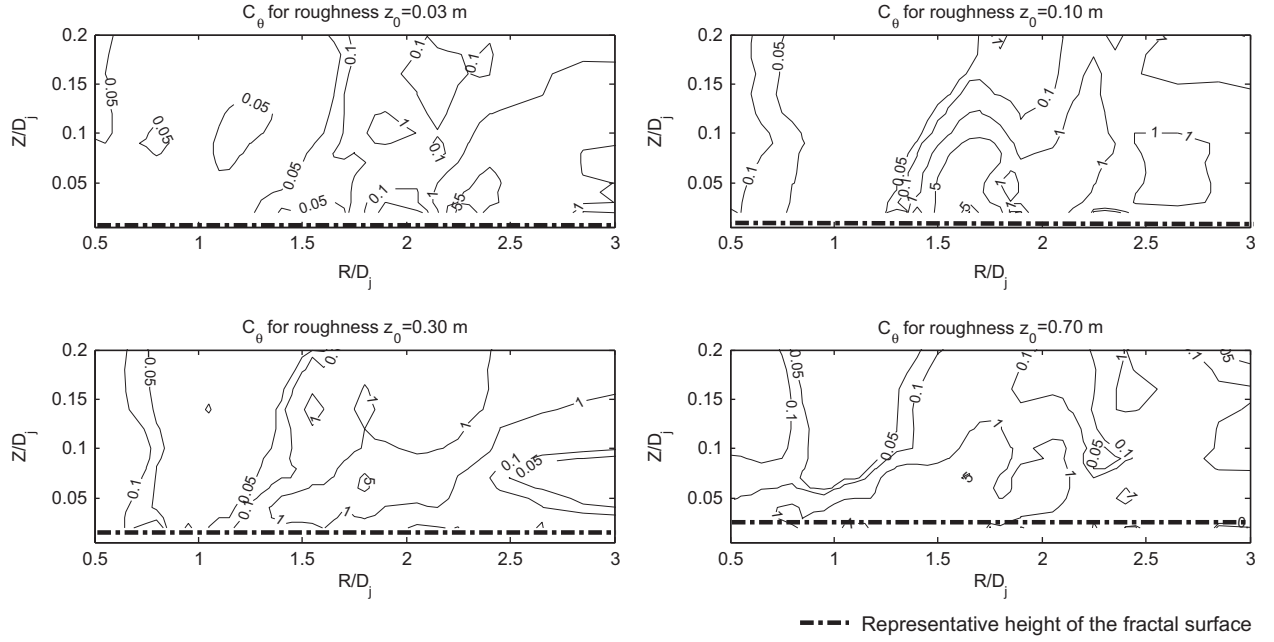


Fig. 23. Coherency decay constant in the circumferential direction C_θ .

method proposed by Davenport (1993). This requires information about the normalized mean velocities, φ_u , turbulent intensities, I_{ur} , turbulent length scales, L_{Db} , peak factor, g_v , that can be easily obtained from the current study.

$$GFF = \frac{\hat{R}}{\bar{R}} = 1 + \frac{2I_{ur}g_v\sqrt{J_{LDb}}}{I_m}$$

$$J_{LDb} = \int \int \varphi_u(n_1)^2 \varphi_u(n_2)^2 i_R(n_1) i_R(n_2) e^{(-L_{st} \cdot |n_1 - n_2| / L_{Db})} dn_1 dn_2$$

$$I_m = \int \varphi_u(n_1)^2 i_R(n_1) dn_1 \quad (19)$$

where: $\hat{R}$, $\bar{R}$ are the peak and the mean responses; J_{LDb} is called the joint acceptance function that depends on the length scales of the downburst turbulence L_{Db} in the direction of the main length of the structure, L_{st} ; $i_R(n)$ is the influence line of the response R ; φ_u is the normalized mean velocity along the structure, where $\varphi_u = U_r(n)/U_{ref}$, and U_{ref} is a reference velocity; n is the local axis of the structure.

Although, the approach described by Eq. (19) involves integrations that need to be evaluated numerically. This approach further can be used to reach closed form expressions for the GFF of different structural responses. For example, Aboshosha and El Damatty (2014) employed such an approach to evaluate the span reduction factor of transmission line conductors subjected to downburst wind and obtained very good matching results with the span reduction factor obtained from a real event (Holmes et al., 2008).

6. Conclusions

Large eddy simulation (LES) of downbursts impinging on various exposure conditions are performed. Ground roughness is simulated by fractal surfaces generated using the random Fourier modes and scaled to produce an aerodynamic roughness, z_0 , equals to 0.03, 0.1, 0.3 and 0.7 m corresponding to open, country side, suburban and urban exposures, respectively. Wind field resulting from the simulations is decomposed into a mean and a turbulent components. Mean component is extracted using a spatial and a temporal averaging. By subtracting the mean

component from the wind field, turbulent component is obtained. Results of the mean wind field show the following:

- Profiles of the peak and instantaneous maximum radial velocity, obtained in the current study for the open exposure, are in a good agreement with the profiles obtained from field measurements, experiments and simulations in the literature.
- Ground roughness is found to affect the profiles of the peak velocities. It is observed with increasing the roughness that the peak velocity decreases and the height where the peak velocity takes place increases, which agree with the trends found in the literature.

Analysis of the turbulent wind field indicates that the turbulent intensity, I , at the locations of the maximum mean velocities ranges between 0.08 and 0.12, 0.08 and 0.16, 0.08 and 0.24, 0.08 and 0.36 for open, country side, suburban and urban exposures, respectively. The average turbulence intensity obtained for the open exposure, $I \sim 0.10$, agrees with the turbulence intensity reported by Holmes et al. (2008) for a real event. Turbulence correlations in the wide frequency range characterized by the turbulent length scales, are investigated and the following are deduced:

- Circumferential length scale, L_θ , reaches up to nine times the jet diameter D_j for the four studied exposures, which indicates that the turbulence associated with downbursts is very well correlated circumferentially.
- Vertical length scales, L_w , ranges between 0.05 and $0.25D_j$, and radial length scales, L_r , ranges between 0.3 and $0.6D_j$ in the zone where mean velocities are maximum. Both are smaller than the circumferential length scales by an order of magnitude.

Peak factor, g_v , of the radial wind fluctuations is found insensitive to the spatial location and generally ranges in between 2 and 2.2. Turbulent correlations in the narrow frequency band characterized by the coherency decay constants are studied in the vertical and the circumferential directions. Decay constant in the vertical direction, C_w , is found to decrease with the increase in the height. The constant has a value in the order of 10 near the ground at the locations of the

maximum mean velocities that is compatible with the normal winds. Decay constant in the circumferential direction, C_θ , is found to be smaller by an order of magnitude than the constant in the vertical direction. This indicates that downburst turbulence is very well correlated in the circumferential direction compared to the vertical direction, which could be unfavorable for long horizontal structures. Finally, the significance of the information presented in this study to calculate the gust front factor (GFF) for structures in general and for transmission lines (TLs) in particular is discussed.

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References

- Aboshosha, H., 2014. Response of Transmission Line Conductors Under Downburst Wind, PhD Thesis, University of Western Ontario.
- Aboshosha, H., El Damatty, A., 2014. Span reduction factor of transmission line conductors under downburst winds. *J. Wind Eng.* 11 (1), 13–22.
- Alahyari, A., Longmire, E.K., 1994. Particle image velocimetry in a variable density flow: application to a dynamically evolving downburst. *Exp. Fluids* 17, 434–440.
- Anderson, J.R., Orf, L.G., Straka, J.M., 1992. A 3-D model system for simulating thunderstorm microburst outflows. *Meteorol. Atmos. Phys.* 49, 125–131.
- Anderson, W., Meneveau, C., 2010. A large-eddy simulation model for boundary-layer flow over surfaces with horizontally resolved but vertically unresolved roughness elements. *Boundary Layer Meteorol.* 137, 397–415.
- Ansys Ltd., 2005. Ansys CFX-solver, Release 10.0: Theory. Canonsburg.
- American Society of Civil Engineers (ASCE), 2010. Guidelines for electrical transmission line structural loading. In: ASCE Manuals and Reports on Engineering Practice, No. 74. New York, NY, USA.
- AS/NZS1170.2, 2011. Standards Australia, Structural design actions. Part 2: Wind actions, Australian-New Zealand Standard.
- Blocken, B., Stathopoulos, T., Carmeliet, J., 2007. CFD simulation of the atmospheric boundary layer: wall function problems. *Atmos. Environ.* 41, 38–252.
- Chay, M.T., Albermani, F., Wilson, R., 2006. Numerical and analytical simulation of downburst wind loads. *Eng. Struct.* 28, 240–254.
- Chay, M.T., Letchford, C.W., 2002. Pressure distributions on a cube in a simulated thunderstorm downburst—Part a: Stationary downburst observations. *J. Wind Eng. Ind. Aerodyn.* 90, 711–732.
- Chay, M., Albermani, F., 2005. Dynamic response of a SDOF system subjected to simulated downburst winds. *Proc., Asia-Pacific Conf. on Wind Engineering (APCWE-VI)*, 1562–1584.
- Chay, M., Albermani, F., Wilson, R., 2006. Numerical and analytical simulation of downburst wind loads. *Eng. Struct.* 28 (2), 240–254.
- Chen, L., Letchford, C.W., 2004a. A deterministic-stochastic hybrid model of downbursts and its impact on a cantilevered structure. *Eng. Struct.* 26 (5), 619–629.
- Chen, L., Letchford, C.W., 2004b. Parametric study on the along wind response of the CAARC building to downbursts in the time domain. *J. Wind Eng. Ind. Aerodyn.* 92 (9), 703–724.
- Chen, L., Letchford, C.W., 2005. Proper orthogonal decomposition of two vertical profiles of full-scale non-stationary correlated downburst wind speeds. *J. Wind Eng. Ind. Aerodyn.* 93, 187–216.
- Choi, E.C.C., Hidayat, F.A., 2002. Dynamic response of structures to thunderstorm winds. *Prog. Struct. Eng. Mech.* 4, 408–416.
- Coccal, O., Dobre, A., Thomas, T.G., Belcher, S.E., 2007. Structure of turbulent flow over regular arrays of cubical roughness. *J. Fluid Mech.* 589, 375–409.
- Darwish, M., Damatty, A., Hangan, H., 2010. Dynamic characteristics of transmission line conductors and behaviour under turbulent downburst loading. *Wind Struct. Int. J.* 13 (4), 327–346.
- Davenport, A.G., 1967. Gust loading factors. *ASCE J. Struct. Div.* 93 (3), 11–34.
- Davenport, A.G., 1993. How can we simplify and generalize wind loads? In: Presented at the Third Asia-Pacific Symposium on Wind Engineering, Keynote Lecture, December 13–15, Hong Kong.
- Engineering Sciences Data Unit (ESDU) 85020, 2001. Characteristics of atmospheric turbulence near the ground. Part II: Single point data for strong winds.
- Fluent Inc., 2005. Fluent 6.2 User's Guide. Fluent Inc., Lebanon.
- Fluent 13.0, 2010. User's Guide. Fluent Inc., Lebanon.
- Franke, J., Hirsch, C., Jensen, A.G., Krüsi, H.W., Schatzmann, M., Westbury, P.S., Miles, S.D., Wisse, J.A., Wright, N.G., 2004. Recommendations on the use of CFD in wind engineering. In: Proceedings of the International Conference on Urban Wind Engineering and Building Aerodynamics, in: van Beeck JPAJ (Ed.), COST Action C14, Impact of Wind and Storm on City Life Built Environment, von Karman Institute, Sint-Genesius-Rode, Belgium, 5–7 May 2004.
- Fujita, T.T., 1990. Downbursts: meteorological features and wind field characteristics. *J. Wind Eng. Ind. Aerodyn.* 36 (1), 75–86.
- Fujita, T.T., 1985. The Downburst Microburst and Macroburst. University of Chicago, Department of Geophysical Sciences p. 128.
- Fujita, T.T., Wakimoto, R.M., 1981. Five scales of airflow associated with a series of downbursts on 16 July 1981. *Mon. Weather Rev.* 109, 1438–1456.
- Germano, M., Piomelli, U., Moin, P., Cabot, W.H., 1991. A dynamic subgrid-scale eddy viscosity model. *Phys. Fluids A* 3 (7), 1760–1765.
- Gant, S.E., 2009. Reliability issues of LES-related approaches in an industrial context. *Flow Turbul. Combust.* 84, 325–335.
- Hadziabdic, M., 2005. LES, RANS and Combined Simulation of Impinging Flows and Heat Transfer (Ph.D Thesis). University of Sarajevo p. 185.
- Hjelmfelt, M.R., 1988. Structure and life cycle of microburst outflows observed in Colorado. *J. Appl. Meteorol.* 27, 900–927.
- Holmes, J.D., 1999. Modeling of extreme thunderstorm winds for wind loading and risk assessment (In: Proceedings of the 10th International Conference on Wind Engineering, Copenhagen, 1999). Balkema Press, Amsterdam, pp. 1409–1455.
- Holmes, J., Hangan, H., Schroeder, J., Letchford, C., Orwig, K., 2008. A forensic study of the Lubbock-Reese downdraft of 2002. *Wind Struct.* 11 (2), 137–152.
- Holmes, J.D., 2008. Recent developments in the specification of wind loads on transmission lines. *J. Wind Eng.* 5 (1), 8–18.
- International Electrotechnical Commission (IEC) 60826, 2003. Design Criteria of Overhead Transmission Lines, Geneva, Switzerland.
- Jeong, J., Hussain, F., 1995. On the identification of a vortex. *J. Fluid Mech.* 285, 69–94.
- Kanda, M., Moriawaki, R., Kasamatsu, F., 2004. Large-eddy simulation of turbulent organized structures within and above explicitly resolved cube arrays. *Boundary Layer Meteorol.* 112, 343–368.
- Katul, G.G., Parlange, M.B., 1995. Analysis of land surface heat fluxes using the orthonormal wavelet approach. *Water Resour. Res.* 31, 2743–2749.
- Kim, J., Hangan, H., 2007. Numerical simulations of impinging jets with application to thunderstorm downbursts. *J. Wind Eng. Ind. Aerodyn.* 95, 279–298.
- Kwon, D., Kareem, A., 2009. Gust-front factor: new framework for wind load effects on structures. *J. Struct. Eng.* 135 (6), 717–732.
- Li, C.Q., 2000. A stochastic model of severe thunderstorms for transmission line design. *Probab. Eng. Mech.* 15, 359–364.
- Lin, W.E., Orf, L.G., Savory, E., Novacco, C., 2007. Proposed large-scale modelling of the transient features of a downburst. *Wind Struct.* 10, 315–346.
- Lundgren, T.S., Yao, J., Mansour, N.N., 1992. Microburst modelling and scaling. *J. Fluids Mech.* 239, 461–488.
- Mason, M., Letchford, C., James, D., 2005. Pulsed wall jet simulation of a stationary thunderstorm downburst, Part A: Physical structure and flow field characterisation. *J. Wind Eng. Ind. Aerodyn.* 93, 557–580.
- Mason, M., Wood, G., 2004. Loading of a very tall building in a simulated downburst wind field. In: Proceedings of the 11th Australasian Wind Engineering Society Workshop, Australasian Wind Engineering Society, Darwin.
- Mason, M., Wood, G., Fletcher, D., 2009. Numerical simulation of downburst winds. *J. Wind Eng. Ind. Aerodyn.* 97, 523–539.
- Mason, M., Fletcher, D., Wood, G., 2010a. Numerical simulation of idealized three-dimensional downburst wind fields. *Eng. Struct.* 32, 3558–3570.
- Mason, M., Wood, G., Fletcher, D., 2010b. Numerical investigation of the influence of topography on simulated downburst wind fields. *J. Wind Eng. Ind. Aerodyn.* 98, 21–33.
- Menter, F.R., Egorov, Y., 2005. A scale adaptive simulation model using two-equation models (43rd AIAA Aerospace Sciences Meeting and Exhibit). American Institute of Aeronautics and Astronautics.
- Nakayama, A., Sakio, K., 2002. Simulation of Flows Over Wavy Rough Boundaries. Center for Turbulence Research, Annual Research Briefs, Stanford University/ NASA Amers Research Center, pp. 313–324.
- Orf, L., Kantor, E., Savory, E., 2012. Simulation of a downburst-producing thunderstorm using a very high-resolution three-dimensional cloud model. *J. Wind Eng. Ind. Aerodyn.* 104–106, 547–557.
- Osegura, R.M., Bowles, R.L., 1988. A simple, analytic 3-dimensional downburst model based on boundary layer stagnation flow. NASA Technical Memorandum, 100632 National Aeronautics and Space Administration.
- Proctor, F.H., 1988. Numerical simulation of an isolated microburst. Part I: Dynamics and structure. *J. Atmos. Sci.* 45, 3137–3160.
- Richards, P.J., Hoxey, R.P., 1993. Appropriate boundary conditions for computational wind engineering models using the *k*- ϵ turbulence model. *J. Wind Eng. Ind. Aerodyn.* 46&47, 145–153.
- Savory, E., Parke, G.A.R., Zeinoddini, M., Toy, N., Disney, P., 2001. Modeling of tornado and microburst-induced wind loading and failure of a lattice transmission tower. *Eng. Struct.* 23 (4), 365–375.
- Selvam, R., Holmes, J.D., 1992. Numerical simulations of thunderstorm downbursts. *J. Wind Eng. Ind. Aerodyn.* 44, 2817–2825.
- Sengupta, A., Sarkar, P.P., 2008. Experimental measurement and numerical simulation of an impinging jet with application to thunderstorm microburst winds. *J. Wind Eng. Ind. Aerodyn.* 96, 345–365.
- Shehata, A.Y., El Damatty, A.A., Savory, E., 2005. Finite element modeling of transmission line under downburst wind loading. *Finite Elem. Anal. Des.* 42 (1), 71–89.
- Simiu, E., Scanlan, R.H., 1996. Wind Effects on Structures, third ed. Wiley, New York, NY.

- Smagorinsky, J., 1963. General circulation experiments with the primitive equations. I. the basic experiment. *Mon. Weather Rev.* 91, 99–164.
- Solari, G., 1993a. Gust buffeting. I: Peak wind velocity and equivalent pressure. *J. Struct. Eng.* 119 (2), 365–382.
- Solari, G., 1993b. Gust buffeting. II: Dynamic along-wind response. *J. Struct. Eng.* 119 (2), 383–397.
- Standards Australia Limited/Standards New Zealand AS/NZS: 7000. 2010. Overhead Line Design—Detailed Procedures, Standard Australia, North Sydney, Australia.
- Teske, M.E., Lewellen, W.S. 1977. Turbulent transport model of a thunderstorm gust front. In: *Proceedings of the 10th Conference on Severe Local Storms*, American Meteorological Society, Omaha.
- Vermeire, B., Orf, L., Savory, E., 2011a. Improved modelling of downburst outflows for wind engineering applications using a cooling source approach. *J. Wind Eng. Ind. Aerodyn.* 99, 801–814.
- Vermeire, B., Orf, L., Savory, E., 2011b. A parametric study of downburst line near-surface outflows. *J. Wind Eng. Ind. Aerodyn.* 99, 226–238.
- Vicroy, D., 1992. Assessment of microburst models for downdraft estimation. *J. Aircr* 29 (6), 1043–1048.
- Wang, W., 2012. An analytical model for mean wind profiles in sparse canopies. *Boundary Layer Meteorol* 142, 383–399.
- Wilson, J.W., Roberts, R.D., Kessinger, C., McCarthy, J., 1984. Microburst wind structure and evaluation of Doppler radar for airport wind shear detection. *J. Clim. Appl. Meteorol.* 23, 898–915.
- Whittingham, H., 1964. Extreme Wind Gust in Australia. Bureau of Meteorology (Melbourne), Department of Science, Bulletin, 46.
- Wood, G.S., Kwok, K.C.S., Motteram, N.A., Fletcher, D.F., 2001. Physical and numerical modelling of thunderstorm downbursts. *J. Wind Eng. Ind. Aerodyn.* 89, 535–552.
- Xie, Z., Coceal, O., Castro, I.P., 2008. Large-eddy simulation of flows over random urban-like obstacles. *Boundary Layer Meteorol.* 129, 1–23.
- Yao, J., Lundgren, T.S., 1996. Experimental investigation of microbursts. *Exp. Fluids* 21, 17–25.
- Zhou, Y., Kareem, A., 2001. Gust loading factor: new model. *J. Struct. Eng.* 127 (2), 168–175.